Identification and Stability of SDF Estimators for Private Equity Cash Flows

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Abstract

We study identification and semiparametric estimation of stochastic discount factors from pooled non-traded cash flows. The framework generalizes the inception-based approach of Driessen et al. (2012) and accommodates dependence induced by overlapping cash-flow streams. We show that horizon choice creates a tradeoff between structural fidelity and finite-sample stability: averaging pricing errors over additional compounding dates can materially stabilize estimation, but generally does so by introducing a Compounding-Horizon Wedge that shifts the population pricing restriction. In private-equity cash-flow panels, the framework yields more stable evidence on broad market exposure in seasoned portfolios than on abnormal performance or richer factor decompositions.

Keywords: Stochastic Discount Factor, Semiparametric Estimation, Spatial Dependence, Generalized Method of Moments (GMM), Private Equity, Illiquid Assets
JEL classification: G12, G24, C14, C58

1 Introduction

Do private equity funds deliver abnormal performance, and how should their systematic risk be measured? These questions are central to a large and growing literature (see Korteweg, 2019, 2023, for comprehensive reviews), yet answering them with cash-flow-based methods confronts a fundamental identification challenge that existing work has not fully addressed. The semiparametric stochastic discount factor (SDF) estimators of Driessen et al. (2012) and Korteweg and Nagel (2016), now standard tools in private-market asset pricing, estimate abnormal performance (α) and factor exposures (β) from pooled cash flows without strong distributional assumptions. But an important practical question remains open: *which* economically meaningful objects can those estimators identify with stability in the sparse, overlapping, and partially NAV-based panels that researchers actually observe?

This paper studies that question by formalizing a tradeoff that practitioners and researchers navigate implicitly in every private-equity SDF estimation. In typical samples, inception-only pricing errors, the foundation of the Driessen et al. (2012) approach, can be highly unstable because the estimating equations are nonlinear and the effective cross-sectional sample size is limited. A natural response is to average pricing errors over additional compounding dates beyond fund inception. But this is not merely a statistical refinement: once pricing errors are evaluated at horizons beyond inception, the estimator generally targets a different population object. We formalize this regularization-like mechanism as the *Compounding-Horizon Wedge* and show that horizon choice creates a tradeoff between finite-sample stabilization and population target shift. Importantly, the simulation evidence below indicates that this estimand shift need not be purely a cost: over an intermediate range of horizons, horizon averaging can improve finite-sample recoverability not only by lowering dispersion, but also by reducing higher-order small-sample bias of the estimator around its horizon-specific target.

Applied to Preqin cash-flow data for 894 buyout and 931 venture-capital funds (North America, USD, vintages 1982–2010), the framework delivers three concrete findings. First, parsimonious single-factor market models yield stable estimates across a wide range of compounding horizons: market betas of approximately 0.8–1.2 for buyout and 1.0–1.9 for venture capital, depending on horizon and weighting scheme. Second, once an α intercept or a second factor is introduced, the estimates become highly unstable, market betas oscillate, sign reversals occur, and asymptotic standard errors become erratic, consistent with weak identification rather than a well-determined multifactor structure. Third, including unseasoned post-2010 vintages with large reported NAV components mechanically inflates estimated market betas (from approximately 1.2 to 2.4 for buyout), underscoring that sample construction is itself an identification choice in private-equity asset pricing.

These findings do not imply that private-equity alpha is zero or that multifactor models are wrong; they indicate that semiparametric alpha estimates from currently available pooled cash-flow panels are weakly identified and highly sensitive to implementation choices. That qualification matters for interpreting both positive and negative findings on private-equity outperformance across the broader literature.

Contributions. The paper makes three contributions.

First, we characterize identification in semiparametric cash-flow-based SDF estimation under flexible horizon choice. We derive a generalized pooled-cash-flow pricing framework that makes explicit how the choice of discounting date affects the estimand, nesting the inception-only criterion of Driessen et al. (2012) as a special case. This formulation reveals a distinction that, to our knowledge, has not been formalized: a compounding horizon may improve finite-sample performance without preserving the original inception-based pricing restriction. The framework also clarifies the relation to Korteweg and Nagel (2016). Their estimator is closely connected but not a literal special case because the order of aggregation differs, so that the two approaches generally converge to different pseudo-true parameter vectors under misspecification.

Second, we formulate the estimator as a nonlinear Least-Mean-Distance (LMD) problem for pooled, non-traded cash flows (Pötscher and Prucha, 1997) and develop an increasing-domain asymptotic framework for inference under dependence induced by overlapping funds and adjacent vintages. This fills a gap in the existing literature: Driessen et al. (2012) use within-portfolio infill asymptotics that do not generate new aggregate-state realizations, while Korteweg and Nagel (2016) rely on nonoverlapping payoff windows that are scarce in typical panels. Our vintage-year increasing-domain device is better aligned with unconditional identification when new vintages bring new macroeconomic realizations. In the empirical application, however, we treat finite-sample simulation evidence and stability diagnostics as the primary basis for interpretation whenever asymptotic approximations appear unreliable.

In that sense, this is an identification and stability paper for private-market asset pricing. *Third,* we provide simulation and empirical evidence on the practical boundaries of semiparametric SDF identification for private equity. We validate the estimator's finite-sample behavior in Monte Carlo experiments calibrated to private-equity-like cash-flow environments and implement hv-block cross-validation diagnostics (Racine, 2000) tailored to overlapping cash-flow dependence. The empirical application demonstrates that, in our sample, current cash-flow

panels support stable inference about broad market exposure in seasoned vintage-year portfolios but not about abnormal performance or richer factor decompositions.

Related literature. The paper contributes to the semiparametric private-equity performance literature that builds on Driessen et al. (2012) and Korteweg and Nagel (2016). The Driessen et al. (2012) estimator has been widely applied to price human capital contracts, venture capital, real estate, and private equity funds (Kroencke et al., 2013; Kim et al., 2015; Farrelly and Stevenson, 2019; Lee et al., 2019; Zhao, 2020; Tausch and Pietz, 2024). The Korteweg and Nagel (2016) framework has found broad application spanning private equity, infrastructure, REITs, CLOs, impact funds, and private debt (Giommetti and Jørgensen, 2021; Gupta and Nieuwerburgh, 2021; Andonov et al., 2021; Riddiough and Wiley, 2022; Boyer et al., 2023; Cordell et al., 2023; Erel et al., 2024; Jeffers et al., 2024; Korteweg and Nagel, 2024; Korteweg et al., 2024). Our LMD framework contributes by providing a loss-function formulation that encompasses Driessen et al. (2012) and contextualizes Korteweg and Nagel (2016) while supplying rigorous increasing-domain asymptotics (drawing on Pötscher and Prucha (1997), Kim and Sun (2011), and Jenish and Prucha (2012)). While Gredil et al. (2020) highlighted that NPV-based measures are biased relative to per-period returns, we explicitly formalize this discrepancy as the Compounding-Horizon Wedge, a concept related to discount-rate risk in secondary markets (Boyer et al., 2023).

More broadly, the paper connects to parametric private-equity models (Franzoni et al., 2012; Ang et al., 2018), the NAV-smoothing literature (Brown et al., 2019; Jenkinson et al., 2020; Brown et al., 2023), credit-market approaches to private-firm valuation (Hüther et al., 2022), and the SDF performance evaluation tradition in public equities (Farnsworth et al., 2002) and hedge funds (Li et al., 2016).

The paper is also complementary to Burnside (2016), who studies identification and inference in linear SDF models with traded excess returns under alternative normalizations. In our pooled non-traded cash-flow setting, the key issue is instead that horizon averaging regularizes the sample criterion while shifting the population pricing restriction.

The remainder of the paper proceeds as follows. Section 2 develops the generalized semiparametric estimator and the corresponding asymptotic inference framework. Section 3 discusses implementation choices and studies finite-sample behavior in simulation. Section 4 presents the empirical application to buyout and venture-capital funds. Section 5 concludes.

2 Methodology

We now formalize a semiparametric framework for SDF estimation with pooled cash flows. Section 2.1 establishes the theoretical asset-pricing conditions and highlights how the choice of discounting date induces the Compounding-Horizon Wedge. Section 2.2 uses this insight to introduce a generalized Least-Mean-Distance estimator that endogenizes the discounting dates. Section 2.3 summarizes the increasing-domain asymptotic benchmark and SHAC inference used to interpret the empirical results under overlapping vintage-year dependence. Section 2.4 demonstrates how this LMD estimator nests the framework of Driessen et al. (2012) as a special case and relates to Korteweg and Nagel (2016).

2.1 Asset pricing for pooled cash flows

Symbol	Description
n	Number of cross-sectional units (funds or portfolios)
V	Number of distinct vintage years
n_v	Number of portfolios in vintage year v
v_i	Vintage year of fund/portfolio i
t_i^0	Inception date of fund i
J_i	Number of deals for fund i
$K_{i,j}$	Number of divestment cash flows for deal j of fund i
$cf_{i,j,t}$	Net cash flow of deal j in fund i at time t
$CF_{i,t}$	Net cash flow of fund i at time t
$P_{\tau,t,i}$	Time-τ price of a single time-t cash flow
$PV_i(\theta)$	Pooled present value
$LCV_{i,\tau}$	Latent continuation value of fund i at date τ
$T, T_i^{\max}$	Final period and maximum fund lifetime
τ	Generic discounting or revaluation date
$\mathcal{F}_\tau$	Information set (filtration) at time τ
$\mathrm{Inv}_{i,j}, \mathrm{Div}_{i,j,k}$	Investment and divestment cash flows
$t_{i,j}^{\mathrm{Inv}}, t_{i,j,k}^{\mathrm{Div}}$	Deal investment and divestment dates
$m_{i,j}^{\mathrm{NFV}}(\cdot)$	NFV-type theoretical moment
Ψ_t , M_t	Multi-period and single-period SDF
$\Psi_{\tau,t}(\theta)$	SDF ratio Ψ_t/Ψ_τ, parameterized by θ
$\theta \in \mathbb{R}^p$	SDF parameter vector (e.g., intercept and factor loadings)
θ_0	True parameter vector
$\delta_{i,j}$	Deal-level pricing error

Table 1: Key notation for asset pricing and pooled cash flows (Section 2.1).

Because private equity funds invest capital over multiple quarters, any fund-level pricing restriction must aggregate deal-level moments across different investment dates. This subsection asks under what conditions such a pooled cash-flow stream satisfies a valid stochastic-discount-factor pricing restriction. The central finding is that the choice of discounting date is not innocuous: discounting at or before the first investment date preserves a clean conditional zero-moment condition (Lemma 1, Part 1), whereas post-investment discounting or future-value compounding generally introduces a timing-related covariance wedge (Parts 2 and 3). The argument proceeds in three steps. First, we state a deal-level decomposition of fund cash flows (Assumption 1) and derive the timing result at the deal level (Lemma 1). Second, we aggregate to the fund level and show that pooled Net Present Value (NPV) moments satisfy a valid zero conditional moment restriction whenever discounting occurs at or before fund inception (Proposition 1). Third, we discuss the implications for Net Future Value (NFV) moments and the interpretation of alpha measured ex post. Throughout, the deal-level structure serves only as a latent accounting device used to illuminate and justify the fund-level restriction; only pooled fund cash flows are required as observable inputs to the estimator.

For fund i, let $CF_{i,t}$ denote the observed net cash flow at time t, and define the fund inception date by $t_i^0 := \min_j t_{i,j}^{\text{Inv}}$. All fund-level variables are constructed from the following deal-level data generation process.

Assumption 1. ***Deal-level decomposition.*** *For each fund i, there exist J_i underlying deals such that:*

1. *Deal j has exactly one negative investment cash flow* $\text{Inv}_{i,j} < 0$ *at time* $t_{i,j}^{\text{Inv}} \in \{0, 1, \ldots, T-1\}$.
2. *Deal j has a finite stream of positive divestment cash flows* $(\text{Div}_{i,j,k})_{k=1,\ldots,K_{i,j}}$ *with* $\text{Div}_{i,j,k} > 0$ *and* $t_{i,j,k}^{\text{Div}} > t_{i,j}^{\text{Inv}}$ *for all k.*
3. *Let $cf_{i,j,t}$ denote the deal-level cash flow of deal j in fund i at time t. Then the observed fund-level cash flow is the cross-deal aggregate*
$$CF_{i,t} = \sum_{j=1}^{J_i} cf_{i,j,t} \qquad \forall i, t.$$
4. *The fund has no cash flows before inception:* $CF_{i,t} = 0$ *for all* $t < t_i^0$.

Under our central asset-pricing restriction, each deal is correctly priced by the stochastic discount factor at its investment date. Formally, the deal-level pricing error satisfies

$$\mathbb{E}\left[\delta_{i,j} \middle| \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] = 0 \quad \forall\, i, j \tag{1}$$

where the deal-level pricing error is defined by

$$\delta_{i,j} := \text{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \frac{\Psi_{t_{i,j,k}^{\text{Div}}}}{\Psi_{t_{i,j}^{\text{Inv}}}} \text{Div}_{i,j,k} \tag{2}$$

and the multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ discounts a time-t cash flow to time 0 (and $\Psi_0 = 1$).

The next result characterizes how this deal-level pricing restriction depends on the choice of discounting or compounding date.

Lemma 1. *Suppose Equation* (1) *holds.*

1. *If* $\tau \le t_{i,j}^{\text{Inv}}$, *then*
$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \middle| \mathcal{F}_\tau\right] = 0. \tag{3}$$
2. *If* $\tau > t_{i,j}^{\text{Inv}}$, *then*
$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \middle| \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\,\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right], \tag{4}$$
which need not equal zero.

3. *Under the same pre-investment timing as Part (1), suppose additionally that the pricing error is compounded forward to a date* $t_{i,j}^{\text{Future}} > t_{i,j}^{\text{Inv}}$. *If* $\tau \leq t_{i,j}^{\text{Inv}} < t_{i,j}^{\text{Future}}$, *then define the NFV-type moment*

$$m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) := \frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_\tau}\,\delta_{i,j}.$$

Its expected value

$$\mathbb{E}\left[m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right] \tag{5}$$

generally contains a conditional covariance term and therefore need not equal zero.

Proof. Consider part (1). Applying the law of iterated expectations,[1] since $\tau \leq t_{i,j}^{\text{Inv}}$, the SDF ratio is known at $t_{i,j}^{\text{Inv}}$ and can be factored out yielding

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right]}_{=0 \text{ by Equation (1)}} \mid \mathcal{F}_\tau\right] = 0.$$

For part (2), since $t_{i,j}^{\text{Inv}} < \tau$, the ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is $\mathcal{F}_\tau$-measurable, so

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right].$$

Since some divestment cash flows of deal j may have been paid by date τ, $\delta_{i,j}$ is partially $\mathcal{F}_\tau$-measurable, and its conditional expectation given $\mathcal{F}_\tau$ generally reflects realized outcomes rather than the zero-mean restriction at $t_{i,j}^{\text{Inv}}$.

For part (3), using the law of iterated expectations with an intermediate conditioning on $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$,

$$\begin{aligned}\mathbb{E}\left[m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) \mid \mathcal{F}_\tau\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] \mid \mathcal{F}_\tau\right] \\ &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right]\end{aligned}$$

where we used the conditional covariance identity and Equation (1). □

Part (3) of Lemma 1 shows that compounding deal cash flows to a future date generally destroys the zero conditional moment restriction because the resulting expression contains a

[1] At the boundary $\tau = t_{i,j}^{\text{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau = 1$ is a trivial constant, so the covariance with $\delta_{i,j}$ is zero and the same argument applies.

covariance between the pricing error and future SDF realizations. We refer to this non-zero expected value as a *timing-related covariance wedge*. Economically, it is the time-τ price of an investment-date covariance term between continuation SDF realizations and the discounted divestment cash flows over the illiquid holding period. At the level of the generalized estimator, this mechanism induces the **Compounding-Horizon Wedge**: once the criterion averages over post-investment or future-value revaluation dates, the timing-related covariance wedge enters the pooled objective and shifts the population target away from the inception-based benchmark.

To clarify the economic mechanism behind the Compounding-Horizon Wedge, let us briefly analyze Equation (5) with only one divestment cash flow per deal ($K_{i,j} = 1$). In this case, the covariance term reads

$$\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}},\ \mathrm{Inv}_{i,j} + \frac{\Psi_{t_{i,j,k}^{\mathrm{Div}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}\mathrm{Div}_{i,j,k} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) = \mathrm{Cov}\left(\prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j}^{\mathrm{Future}}} M_u,\ \mathrm{Div}_{i,j,k}\prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j,k}^{\mathrm{Div}}} M_u \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right)$$

Thus, for $K_{i,j} = 1$, the covariance is naturally expected to be largest when $t_{i,j}^{\mathrm{Future}}$ is equal to $t_{i,j,k}^{\mathrm{Div}}$, since the continuation SDF and the discounted divestment payoff then load on the same SDF path. This ranking is not automatic, however, and requires additional restrictions on the post-divestment SDF dynamics (e.g, $\Psi_{t_{i,j}^{\mathrm{Future}}}/\Psi_{t_{i,j}^{\mathrm{Div}}}$ being conditionally mean-independent of $\mathcal{F}_{t_{i,j}^{\mathrm{Div}}}$ given $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$).

To conclude our analysis of Part 3 of Lemma 1, NFV-type moments are inappropriate as primitive ex-ante pricing moments unless the timing-related covariance wedge is modeled explicitly. They can still be informative for ex-post benchmarking, but the resulting performance measure must then be interpreted as a composite object that mixes true abnormal performance (skill) with systematic exposure to the states in which realization and discounting occur.

Proposition 1. ***Price of a pooled cash-flow stream at or before fund inception.*** *If $\tau \le t_i^0$, then*

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,\middle|\,\mathcal{F}_\tau\right] = 0. \tag{6}$$

Proof. If $\tau \le t_i^0$, then $\tau \le t_{i,j}^{\mathrm{Inv}}$ for every deal j in fund i. Using the deal-level decomposition of the pooled cash-flow stream and Equation (2),

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,\middle|\,\mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J_i}\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right]. \tag{7}$$

Applying Equation (3) term-by-term yields

$$\mathbb{E}\left[\sum_{j=1}^{J_i}\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right] = 0. \tag{8}$$

□

Proposition 1 provides the clean benchmark for identification: pooled NPV moments are theoretically well founded when discounting occurs at or before fund inception. Once $\tau > t_i^0$, some underlying deals are active or already partially realized, so the corresponding pooled pricing error incorporates the Compounding-Horizon Wedge. The choice of discounting dates is therefore not innocuous. Section 2.2 formalizes this choice by defining an estimator that operates entirely on pooled fund cash flows (the deal-level decomposition having served its purpose as an analytical device) and explicitly specifies which discounting dates are used. When it restricts to dates $\tau \leq t_i^0$, the pooled fund-level pricing error inherits a valid zero conditional moment. When it includes dates $\tau > t_i^0$, it generally targets a tilted criterion that embeds the Compounding-Horizon Wedge characterized by Equation (5).

Critically, the Compounding-Horizon Wedge is not solely relevant for ex-post benchmarking. It also fundamentally drives the large-sample and small-sample properties of our generalized LMD estimator. As we demonstrate in later sections, averaging pricing errors over extended horizons deliberately introduces wedge-bearing discounting dates that smooth the nonlinear sample criterion and can materially improve finite-sample behavior, even though doing so shifts the population target.

2.2 Least-Mean-Distance estimator

Symbol	Description
$\epsilon_{\tau,i}$	Fund pricing error at discounting date τ
$\bar{\epsilon}_i, \bar{\epsilon}_i^{(H)}$	Weighted average pricing error (generic and horizon-specific)
H	Horizon parameter
$\mathcal{T}_i, \mathcal{T}_{i,H}$	Set of discounting dates (generic and horizon-specific)
w_i	Weighting factor for fund i
$L(\cdot)$	Loss function (e.g., $L(x) = x^2$)
$Q_H(\theta)$	Horizon-specific population objective function
$Q_n(\theta), Q_{n,H}(\theta)$	Sample objective function (generic and horizon-specific)
$s_i^{(H)}(\theta)$	Unit score contribution
$d_{i,j}$	Economic distance between units i and j
$K(\cdot), k_{i,j}$	Symmetric SHAC kernel function and generic weights
b_V	SHAC bandwidth parameter
$\hat{\theta}, \hat{\theta}_H$	LMD Estimator (generic and horizon-specific targets)
$\theta_H^\dagger$	Horizon-specific pseudo-true parameter vector
$\mathcal{H}_H$	Hessian of population objective at $\theta_H^\dagger$
Λ_H	Long-run covariance matrix of the score
Σ_H	Asymptotic covariance matrix of $\hat{\theta}_H$

Table 2: Key notation for the LMD estimator and asymptotic benchmark (Sections 2.2–2.3).

In this subsection, we define a generalized fund-level criterion that averages pricing errors across a user-specified set of discounting dates $\mathcal{T}_i$. This nests the inception-only NPV criterion

of Driessen et al. (2012) as a special case and allows us to study how horizon averaging trades structural validity against finite-sample regularization. Throughout, only pooled fund cash flows are observed, whereas the deal-level decomposition from Section 2.1 remains a latent theoretical device.

The stochastic discount factor $\Psi_{\tau,t} = \Psi_t/\Psi_\tau$ is used to calculate the time-τ price $P_{\tau,t,i}$ of a **single** time-t cash flow of any given PE fund i

$$P_{\tau,t,i} := \Psi_{\tau,t} \cdot CF_{i,t} = \frac{\Psi_t}{\Psi_\tau} \cdot CF_{i,t} \qquad \forall \quad \tau, t, i \tag{9}$$

with again multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ and $\Psi_0 = 1$. As SDFs are commonly parameterized by a vector $\theta \in \mathbb{R}^p$, i.e., $\Psi_{\tau,t} \equiv \Psi_{\tau,t}(\theta)$, our goal is to find an estimation method for the true parameter vector θ_0.

For each fund i and all points τ within a common fund lifetime, the empirical pricing error $\epsilon_{\tau,i}(\theta)$ of **all** realized-to-date fund cash flows is calculated as a time-τ revaluation of the entire cash-flow stream

$$\epsilon_{\tau,i}(\theta) := \sum_{t=0}^{T} P_{\tau,t,i}(\theta) \qquad \forall \quad \tau, i \tag{10}$$

Note that for $t < \tau$, the terms $\frac{\Psi_t}{\Psi_\tau} CF_{i,t}$ are compounded past cash flows, not discounted future cash flows, so the term "present value" is no longer accurate for $\tau > t_i^0$. For $\tau > t_i^0$, $\epsilon_{\tau,i}(\theta)$ is not a continuation value but a time-τ revaluation of the entire realized cash-flow stream, and therefore generally inherits the Compounding-Horizon Wedge characterized in Section 2.1. This time-τ revaluation $\epsilon_{\tau,i}(\theta)$ must be sharply distinguished from the latent continuation value $LCV_{i,\tau}(\theta) := \mathbb{E}[\sum_{t=\tau}^{T}(\Psi_t/\Psi_\tau)CF_{i,t} \mid \mathcal{F}_\tau]$, which conditions on information at τ and is not directly observed.

To analyze the effect of different discounting dates τ on the bias and variance of the estimated SDF parameters, we define the generalized criterion $\bar{\epsilon}_i(\theta)$ that averages pricing errors over the set $\mathcal{T}_i$:

$$\bar{\epsilon}_i(\theta) = w_i \cdot \frac{1}{\text{card}(\mathcal{T}_i)} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i}(\theta) \qquad \forall \quad i \tag{11}$$

where w_i normalizes fund-level pricing errors before the loss is applied; it is not an observation weight in the outer sample average. Depending on the application, w_i may normalize by invested capital, by aggregate vintage-year capital, by a macroeconomic deflator, or be set to one when cash flows are already expressed in comparable scale units.

The role of $\mathcal{T}_i$ The set $\mathcal{T}_i$ controls which discounting dates enter the averaged pricing error $\bar{\epsilon}_i(\theta)$ in Equation (11) (see Figure 1). Averaging over multiple dates in $\mathcal{T}_i$ was originally motivated by the "exploding alpha" problem described by Driessen et al. (2012): when the initial cash flow is very small, an arbitrarily large alpha can discount all subsequent cash flows close to zero, rendering the beta factors irrelevant and distorting estimation. Thus, their inception-only objective function is not globally convex in finite samples: the multiplicative SDF parameterization permits extreme parameter regions in which the discount factors drive all NPV_i close to zero, creating spurious near-minima that can prevent reliable identification of

α. Averaging over $\mathcal{T}_i$ mitigates this degeneracy (see Figure 2). The relevant horizon family is parameterized as

$$\mathcal{T}_{i,H} = \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\}, \qquad H \geq 0. \tag{12}$$

This formulation encompasses two economically distinct cases:

- $H = 0$: If $\mathcal{T}_i = \{t_i^0\}$, the estimator uses an inception-only NPV criterion, is anchored in the pooled pricing restriction from Section 2.1, and targets the structural parameter θ_0.
- $H > 0$: If $\mathcal{T}_i = \mathcal{T}_{i,H}$ with $H > 0$, the estimator is a generalized horizon-averaged criterion and targets the pseudo-true parameter $\theta_H^\dagger$. It acts as a finite-sample regularization device by smoothing the sample criterion, but it does so by introducing the Compounding-Horizon Wedge and therefore changing the estimand.

Larger values of the horizon H offer more regularization, but a larger departure from the structural benchmark with zero expected value.

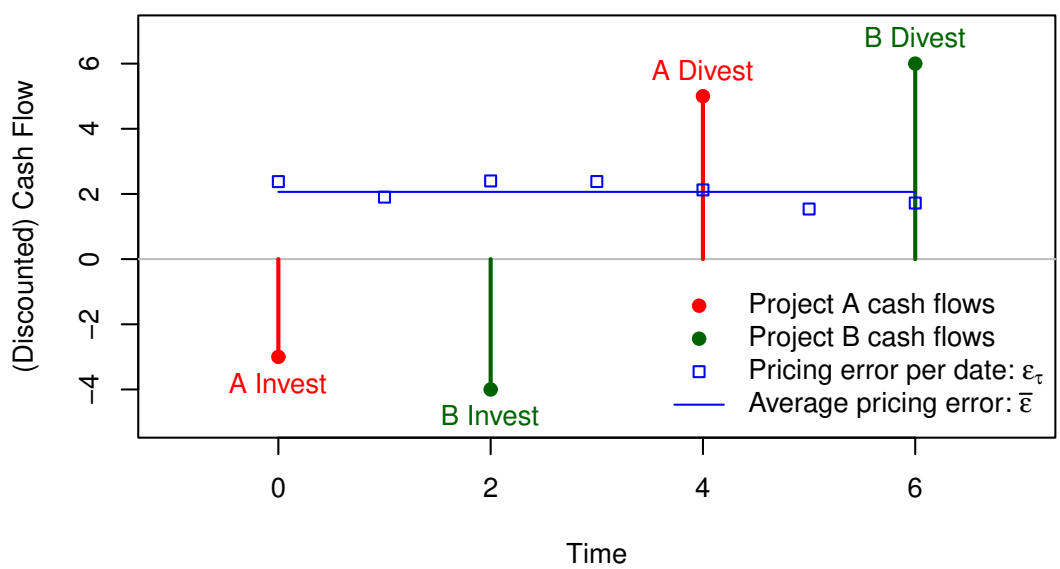

Figure 1: How to calculate and interpret the average pricing error introduced by Equation (11): The time index t is relevant for the net cash flows (black dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation (11) (solid blue line). In this example, $\text{card}(\mathcal{T}_i) = 7$, i.e., the number of blue boxes.

For $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, the estimator targets the horizon-specific pseudo-true parameter $\theta_H^\dagger$, which coincides with the structural parameter θ_0 only when $H = 0$. For $H > 0$, the estimator trades structural fidelity against finite-sample regularization by averaging pricing errors over discounting dates that may incorporate the Compounding-Horizon Wedge from Section 2.1.

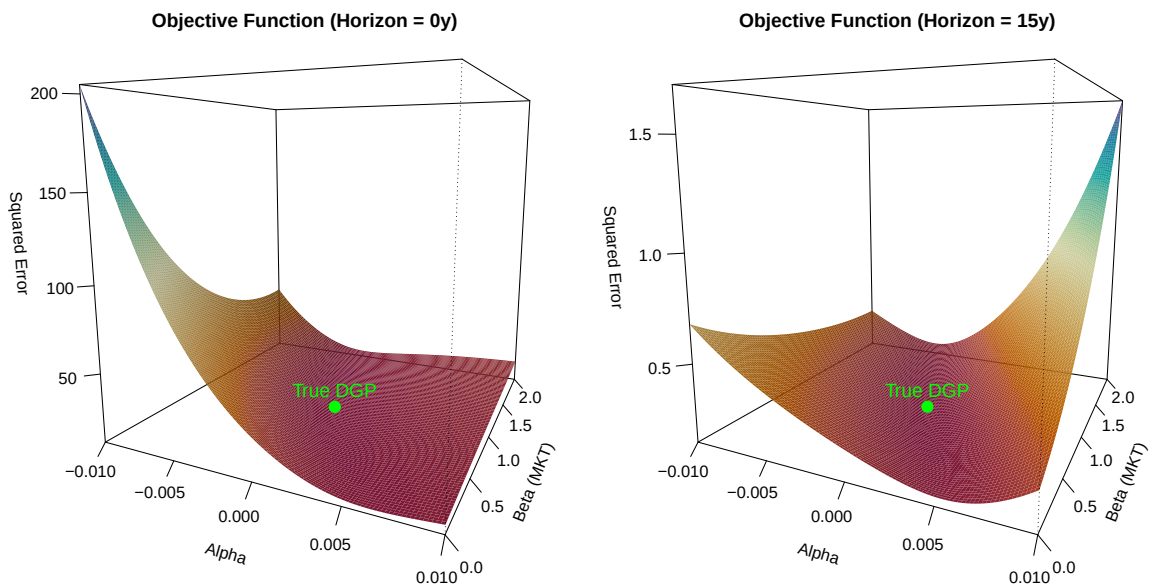

Figure 2: The figure plots the sample objective function $Q_{n,H}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \bar{\epsilon}_i(\theta)^2$ over a two-dimensional grid of SDF parameters $\alpha \in [-0.01, 0.01]$ and $\beta_{\text{MKT}} \in [0, 2]$, using simulated private-equity cash flows (from the base-case DGP of Section 3.5) with true parameters $\alpha_0 = 0$ and $\beta_{\text{MKT},0} = 1$. The left panel uses the inception-only criterion ($H = 0$), where $\mathcal{T}_i = t_i^0$ and each fund's pricing error $\bar{\epsilon}_i(\theta)$ reduces to a single unscaled net present value $PV_i(\theta) = \sum_t \Psi_{t_i^0,t}(\theta) CF_{i,t}$. The right panel uses horizon averaging with $H = 15$ years, so that $\bar{\epsilon}_i(\theta)$ averages pricing errors over $\text{card}(\mathcal{T}_{i,H})$ discounting dates as defined in Equation (11). In each panel, the green dot marks the true DGP parameter vector $(\alpha_0, \beta_{\text{MKT},0}) = (0, 1)$. The left panel illustrates the "exploding alpha" degeneracy: extreme values of α drive the multiplicative discount factors $\Psi_{\tau,t}(\theta)$ toward zero for all funds, collapsing the squared pricing errors to near-zero and creating a ridge of spurious near-minima along the α axis that prevents reliable identification. The right panel shows that horizon averaging over $\mathcal{T}_{i,H}$ regularizes the objective surface, restoring a well-identified minimum near the true parameter values.

Restriction used for identification and asymptotic theory. In the consistency analysis below, we specialize the generalized criterion to $\mathcal{T}_i = \{t_i^0\}$. Under Proposition 1, this yields a valid pooled fund-level zero-moment restriction at the structural parameter θ_0. This restriction guarantees that $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$ under the true parameter. For the quadratic loss $L(x) = x^2$, the population objective $Q_0(\theta) = -\mathbb{E}[\bar{\epsilon}_i(\theta)^2] = -\mathrm{Var}[\bar{\epsilon}_i(\theta)] - (\mathbb{E}[\bar{\epsilon}_i(\theta)])^2$ involves both variance and bias terms, so consistency requires that θ_0 uniquely maximizes this composite criterion (see Proposition 2).

LMD estimator. Because our estimator is defined through the optimization of a sample criterion, it is an extremum estimator in the sense of Newey and McFadden (1994, Eq. 1.1), with $\hat{\theta} = \arg\max_{\theta\in\Theta} Q_n(\theta)$. Concretely, our Least Mean Distance (LMD) estimator (Pötscher and Prucha, 1997, Equation 7.1) minimizes the average loss of $\bar{\epsilon}_i(\theta)$. For consistency with the extremum-estimator convention, we will technically maximize the average negative loss of $\bar{\epsilon}_i(\theta)$, which is equivalent.

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \ \frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big) \equiv \arg\max_{\theta\in\Theta} \ Q_n(\theta) \tag{13}$$

with

$$Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big)$$

where $L : \mathbb{R}^1 \to \mathbb{R}^1$ denotes a loss function, e.g., $L(x) = (x-0)^2$ in the case of nonlinear least squares. Throughout the paper, the weighted average fund pricing error $\bar{\epsilon}_i \equiv \bar{\epsilon}_i(\theta)$ is regarded as a nonlinear random function of the SDF parameter θ.

2.3 Asymptotic benchmark and SHAC inference

The asymptotic theory plays a supporting role in the paper. It provides a large-sample benchmark for the horizon-indexed criterion actually being estimated. It also clarifies the population object associated with each horizon and motivates the dependence-robust covariance estimator used later in the empirical analysis. Because the available private-equity panels remain small and heavily overlapping, we do not treat asymptotic approximations as the primary source of empirical credibility in finite samples.

The large-sample analysis is stated for the horizon-indexed estimator

$$\hat{\theta}_H = \arg\max_{\theta\in\Theta} Q_{n,H}(\theta), \qquad Q_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n} L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \tag{14}$$

where $\bar{\epsilon}_i^{(H)}(\theta)$ denotes the horizon-averaged pricing error based on $\mathcal{T}_{i,H}$ from Section 2.2. Throughout this subsection, the horizon parameter $H \geq 0$ is fixed while $n \to \infty$. The notation makes explicit that both the estimator and its probability limit generally depend on H.

Define the horizon-specific population objective by

$$Q_H(\theta) := \mathbb{E}\big[Q_{n,H}(\theta)\big] = -\mathbb{E}\Big[L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big)\Big], \tag{15}$$

and let the corresponding population target be

$$\theta_H^\dagger := \arg\max_{\theta\in\Theta} Q_H(\theta). \tag{16}$$

The asymptotic benchmark is intentionally formulated around $\theta_H^\dagger$, not around θ_0. For $H = 0$, the pooled inception-only criterion coincides with the structural pricing restriction of Section 2.1; the benchmark target is structural when the inception-only population criterion is uniquely maximized at θ_0. For $H > 0$, the criterion generally targets a pseudo-true parameter $\theta_H^\dagger \neq \theta_0$, because averaging over $\tau > t_i^0$ may embed the Compounding-Horizon Wedge at the population level.

We use increasing-domain asymptotics indexed by vintage years, with local overlap dependence and bounded within-vintage clustering; the full regularity conditions are stated in Online Appendix C. Under those conditions, the estimator fits the semiparametric extremum framework of Pötscher and Prucha (1997) and Newey and McFadden (1994), while the required uniform law of large numbers and random-field central limit theorem follow from Jenish and Prucha (2009, 2012).

Proposition 2 (Consistency around the horizon-specific target)**.** *Under the regularity conditions stated in Online Appendix C,*

$$\hat\theta_H \xrightarrow{p} \theta_H^\dagger \qquad \text{as } n \to \infty.$$

This is the horizon-indexed analogue of the standard extremum-estimator consistency result of Newey and McFadden (1994). For $H = 0$, the probability limit coincides with the structural parameter θ_0 when the inception-only population criterion is uniquely maximized at θ_0; for $H > 0$, consistency is around the horizon-specific pseudo-target $\theta_H^\dagger$.

Proposition 3 (Asymptotic normality around the horizon-specific target)**.** *Under the regularity conditions stated in Online Appendix C,*

$$\sqrt{n}\left(\hat\theta_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_H), \qquad \Sigma_H = \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}.$$

This combines the extremum-estimator expansion of Newey and McFadden (1994) with the increasing-domain random-field CLT of Jenish and Prucha (2012). As with consistency, the special case $H = 0$ is centered at the structural parameter θ_0 only under the additional structural-identification condition that the inception-only population criterion is uniquely maximized at θ_0; for $H > 0$ the asymptotic center is the pseudo-target $\theta_H^\dagger$.

For large-sample inference, let $s_i^{(H)}(\theta) := \nabla_\theta L(\bar\epsilon_i^{(H)}(\theta))$ denote the score contribution and $\hat{\mathcal{H}}_H := \nabla_{\theta\theta} Q_{n,H}(\hat\theta_H)$ the sample Hessian. We estimate the long-run score covariance with the spatial HAC estimator of Kim and Sun (2011) adapted to the vintage-year metric:

$$\hat\Lambda_H = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} k_{i,j}\, s_i^{(H)}(\hat\theta_H) s_j^{(H)}(\hat\theta_H)^\top, \qquad k_{i,j} = K\left(\frac{d_{i,j}}{b_V}\right), \tag{17}$$

where $d_{i,j}$ is the vintage-year distance metric and b_V is the SHAC bandwidth; the full assumptions and implementation details are given in Online Appendix C. The resulting asymptotic covariance matrix is

$$\widehat{\text{Avar}}(\hat{\theta}_H) = \frac{1}{n}\hat{\mathcal{H}}_H^{-1}\hat{\Lambda}_H\hat{\mathcal{H}}_H^{-1}. \tag{18}$$

Asymptotic standard errors then follow in the usual way $\text{SE}(\hat{\theta}_H) = \sqrt{\text{diag}\left[\widehat{\text{Avar}}(\hat{\theta}_H)\right]}$.

When $H > 0$, inference based on Equation (18) pertains to the pseudo-true parameter $\theta_H^\dagger$; it therefore serves as a large-sample benchmark for stability diagnostics and model comparison, not as a direct structural test about θ_0.

Even when finite-sample inference is weak, this asymptotic benchmark still matters for interpretation. It defines the population object associated with each horizon-indexed criterion, makes the distinction between the structural parameter θ_0 and the horizon-specific pseudo-target $\theta_H^\dagger$ precise, and motivates the SHAC covariance estimator as the appropriate large-sample reference under overlap dependence. In the empirical sections, we use this benchmark to interpret, rather than override, the simulation and cross-validation evidence.

2.4 Comparison to similar estimators

Our Least-Mean-Distance (LMD) estimator belongs to the broad class of semiparametric non-linear extremum estimators in the sense of Pötscher and Prucha (1997). Its main advantage in the present setting is that it accommodates pooled fund- or portfolio-level pricing errors, flexible horizon averaging through $\mathcal{T}_{i,H}$, and increasing-domain asymptotics indexed by vintage years, all within a comparatively transparent objective function.

This section compares it to three nearby contributions, each along a different margin. Relative to Driessen et al. (2012), our criterion nests the inception-only objective exactly, but relies on a different asymptotic device. Relative to Driessen et al. (2009), our horizon averaging addresses the same stabilization problem as ratio or log-ratio transformations, but changes the bias channel differently. Relative to Korteweg and Nagel (2016), we share the discounted-cash-flow pricing logic, but differ in aggregation and in the asymptotic source of identifying variation.

2.4.1 Connection to Driessen et al. (2012)

The connection to Driessen et al. (2012) is the closest at the objective-function level. Under the inception-only restriction, equal normalization, and quadratic loss, the LMD criterion coincides with the identity-weighted one-step cross-sectional GMM objective used in their framework. Recall that Driessen et al. (2012) estimate $(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{i=1}^n [NPV_i(\alpha, \beta)]^2$, where NPV_i discounts all cash flows of portfolio i to fund inception. To bridge the gap, let us rewrite NPV_i in our more general SDF notation as $PV_i(\theta) = \sum_{t\in\{t_i^0, t_i^0+1,\dots\}} \Psi_{t_i^0,t}(\theta)CF_{i,t}$, where t_i^0 denotes the fund inception date as defined in Section 2.1.

Proposition 4 (Connection to the Driessen et al. (2012) objective)**.** *Suppose that:*

1. $\mathcal{T}_i = \{t_i^0\}$ *for all i,*

2. $w_i = 1$ *for all i,*

3. $L(x) = x^2$,

4. *unit i is a vintage-year portfolio.*

Then the LMD estimator solves

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 ,$$

which is equivalent to the identity-weighted cross-sectional GMM objective in Driessen et al. (2012).

Proof. Under $\mathcal{T}_i = \{t_i^0\}$, the averaged pricing error reduces to the inception-based pooled net present value

$$\bar{\epsilon}_i(\theta) = \epsilon_{t_i^0,i}(\theta) = \sum_{t=0}^{T} \frac{\Psi_t(\theta)}{\Psi_{t_i^0}(\theta)} CF_{i,t} = PV_i(\theta).$$

With $w_i = 1$ and $L(x) = x^2$, Equation (13) becomes

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 .$$

Writing the vector of pooled pricing errors as $e(\theta) = (PV_1(\theta), \ldots, PV_n(\theta))^\top$, the objective is proportional to $e(\theta)^\top e(\theta)$, which is the identity-weighted one-step GMM criterion. □

Driessen et al. (2012) justify this inception-based estimator under a different asymptotic device than the one used here. Their Theorem 1 fixes the number of vintage-year portfolios and lets the number of underlying projects or funds within each portfolio diverge, yielding consistency for the inception-only estimator under their maintained model (Driessen et al., 2012, p. 520). Economically, this portfolio-infill asymptotic sequence averages out within-portfolio idiosyncratic noise, but it does not by itself create new realizations of aggregate states. Relative to that setup, our framework preserves the exact $H = 0$ nesting result while letting the vintage-year domain itself expand, which is better aligned with unconditional identification, horizon comparisons, and overlap dependence when new vintages bring new realizations of macroeconomic conditions.

2.4.2 Connection to Driessen et al. (2009)

The earlier working-paper version, Driessen et al. (2009), provides the closest conceptual precursor to our NFV approach. Starting from the same inception-based NPV criterion as Driessen et al. (2012), they show that the raw sum of squared NPVs is not globally convex in finite samples (the "exploding alpha" problem mentioned before) and stabilize estimation by transforming the NPV relation itself. Their two proposed "GMM-style" criteria are a log-transformed specification based on the difference between the log present values of investments

and dividends and a PME-style ratio specification akin to Kaplan and Schoar (2005) that minimizes the distance between the ratio of these present values and one. This is the main conceptual connection to our framework: both approaches use transformed-NPV criteria to obtain a better-behaved objective, but the source of the resulting bias is different.

To make the asymptotic logic explicit without introducing additional counting notation, let $\overline{PV}^D(\theta)$ and $\overline{PV}^I(\theta)$ denote the discounted portfolio averages of dividends and investments within a generic vintage-year portfolio, and let $\overline{NPV}(\theta) = \overline{PV}^D(\theta) - \overline{PV}^I(\theta)$. Under the working paper's asymptotic scheme, the number of funds inside each fixed vintage-year portfolio diverges, so the PME transformation inherits the same population zero set as the inception-NPV criterion:

$$\frac{\mathbb{E}[\overline{PV}^D(\theta_0)]}{\mathbb{E}[\overline{PV}^I(\theta_0)]} - 1 = \frac{\mathbb{E}[\overline{NPV}(\theta_0)]}{\mathbb{E}[\overline{PV}^I(\theta_0)]}.$$

Hence, when $\mathbb{E}[\overline{NPV}(\theta_0)] = 0$ and $\mathbb{E}[\overline{PV}^I(\theta_0)] > 0$, the transformed and untransformed portfolio criteria are asymptotically equivalent. The cost of stabilization is therefore not a different asymptotic target, but a finite-sample nonlinearity bias: Driessen et al. (2009) find a downward bias for α under the log transformation and an upward bias under the ratio specification.

The contrast to our framework is that their transformation is applied after within-portfolio averaging. If the ratio transformation were treated as primitive at the unit level before aggregation, the target would change. For a generic unit i with $NPV_i(\theta) = PV_i^D(\theta) - PV_i^I(\theta)$, one obtains at θ_0

$$\mathbb{E}\left[\frac{PV_i^D(\theta_0)}{PV_i^I(\theta_0)}\right] - 1 = \mathbb{E}\left[\frac{NPV_i(\theta_0)}{PV_i^I(\theta_0)}\right] = \mathrm{Cov}\left(NPV_i(\theta_0), \frac{1}{PV_i^I(\theta_0)}\right),$$

where the last equality uses $\mathbb{E}[NPV_i(\theta_0)] = 0$. This expectation of a random ratio is not the ratio-of-expectations moment justified by Driessen et al. (2009). The Driessen et al. (2009) construction therefore stabilizes estimation while preserving the same large-sample target under portfolio-infill asymptotics, whereas in our setup changing H changes the set of discounting dates and hence the population criterion itself. Different horizons can therefore target different pseudo-true parameters $\theta_H^\dagger$ even asymptotically. A practical advantage of our horizon-based approach is that it operates directly on net cash-flow streams. The two alternatives in Driessen et al. (2009), by contrast, require separately observed and accurately labeled investment and dividend cash flows.

2.4.3 Connection to Korteweg and Nagel (2016)

Our framework is close to Korteweg and Nagel (2016) at the level of primitive discounted cash-flow moments, but it does not literally nest their estimator. If the cross-sectional unit is allowed to be an arbitrary pooled portfolio or calendar-time window, our notation can represent the same private-equity cash-flow moment building blocks that enter their GMM setup. In that sense, the connection is economically close.

The main difference is the sample objective. Our LMD estimator applies the loss before aggregation, minimizing an average of unit-level losses, whereas Korteweg and Nagel (2016)

apply a GMM quadratic form to an aggregated mean moment vector. While both approaches target the true parameter θ_0 under correct model specification, this distinction becomes economically meaningful under misspecification. Because the *average of losses* applies a different implicit asset weighting than the *loss of an average*, the two approaches will generally converge to different pseudo-true parameter vectors when the model is misspecified, a pervasive feature of asset pricing models. In addition, their empirical implementation assigns zero weight to the private-equity pricing-error block during estimation, so the SDF is pinned down by public-equity and T-bill moments and the PE pricing error is evaluated only afterwards. Accordingly, Korteweg and Nagel (2016) are best viewed as a closely related aggregated-moment GMM analogue rather than as a special case of Equation (13).

The asymptotic difference mirrors this design choice. The Korteweg and Nagel (2016) approach is natural when identification comes from repeated aggregate-state realizations: enlarging the cross section within a common payoff window averages out idiosyncratic fund noise, but not the common SDF realization itself. Their asymptotic logic therefore relies on additional nonoverlapping payoff windows rather than on cross-sectional replication alone (Korteweg and Nagel, 2016, p. 1447). Our framework instead uses increasing-domain asymptotics indexed by vintage years, with bounded within-vintage counts and bounded fund lifetimes, so that new vintages contribute new macroeconomic realizations while dependence remains local.

2.4.4 Positioning of the present estimator

The present estimator is best viewed as a synthesis tailored to the current application. It combines the exact inception-only objective link to Driessen et al. (2012), the stabilization concern emphasized in Driessen et al. (2009), and the Korteweg and Nagel (2016) insight that identification requires new aggregate-state realizations rather than cross-sectional replication alone. Operationally, these elements enter a pooled-unit nonlinear least-squares framework with explicit horizon control.

The practical distinction is which form of sample expansion supplies new identifying information. If a researcher mainly gains more funds inside already-observed vintages, the Driessen et al. (2012) infill logic is the natural approximation. If the sample mainly lengthens in calendar time and thereby adds nonoverlapping payoff windows, the Korteweg and Nagel (2016) logic is closer to the underlying information structure. If the panel broadens chiefly by observing additional vintages while per-vintage counts remain modest, our increasing-domain framework is the more natural approximation. Table 3 summarizes the main distinctions between Driessen et al. (2012), Korteweg and Nagel (2016), and our approach.

3 Implementation choices and simulation

In this section and the next, we move from the general methodology to our empirical application. First, we describe the data and establish our sample restriction based on terminal cash-flow measurement concerns. Second, we detail the empirical implementation choices that anchor the general estimator from Section 2.2. Third, we use a simulation study to validate and calibrate these choices. Finally, Section 4 presents the resulting empirical estimates for Buyout and

	Driessen et al. (2012)	Korteweg and Nagel (2016)	Our approach
Estimator type	Cross-sectional NLS (one-step GMM)	Time-series-like GMM	Nonlinear Least-Mean-Distance
Cash flows priced	PE fund cash flows	PE payoffs + public benchmark moments	PE fund cash flows
Discounting dates	Fund inception only	Fund inception only	Flexible via $\mathcal{T}_i$
What grows	Funds/projects within fixed vintage portfolios	Calendar span with new nonoverlapping payoff windows	Number of vintage years with bounded n_v
What is averaged out	Within-portfolio idiosyncratic fund/project noise	Sampling noise in aggregated SDF-return moments across windows	Short-range dependence across vintages
More data help when	Existing vintages receive more funds	The sample adds new nonoverlapping time windows	The sample adds new vintages, not merely more funds per vintage
Inference	Bootstrap	Spatial HAC	Spatial HAC
Cross-sectional unit	Vintage year portfolio	Single payoff series	Both (tested empirically)
SDF specification	Linear factor model	Exponentially affine	Both (tested empirically)

Table 3: Comparison of SDF estimation frameworks for private equity.

Venture Capital funds.

3.1 Private data

We use the Preqin cash flow dataset as of 14th April 2022, which is well known in the academic private equity literature (Harris et al., 2014; Brown et al., 2015; Korteweg and Nagel, 2016; Ang et al., 2018). To keep the empirical analysis clear and concise, we filter the dataset for Buyout (BO) and Venture Capital (VC) funds. We only keep funds with fund currency US Dollar (USD) and region equal to North America.

For this subset of BO and VC funds, we extract all (i) equal-weighted (hereafter EW) and (ii) value-weighted (hereafter FW, weighted by fund size) cash flow series (cf. Figures 3 and 4). In the overall dataset, the minimum vintage year is 1982, and the maximum is 2021. We initially retain all vintage years, including recent ones. Section 3.3 assesses the impact of unseasoned fund Net Asset Values (NAVs) on our empirical estimation results. On this basis, we decide how many recent vintage years to remove to reduce reliance on unrealized NAVs. For the vintage years we keep, all liquidated and non-liquidated funds enter our estimation. For non-liquidated funds, we treat the latest NAV as final cash flow.

Buyout sample. The Buyout (North America) sample contains 894 distinct funds spreading over 36 vintage years. The vintage year distribution is as follows: 1985 (2), 1986 (3), 1987

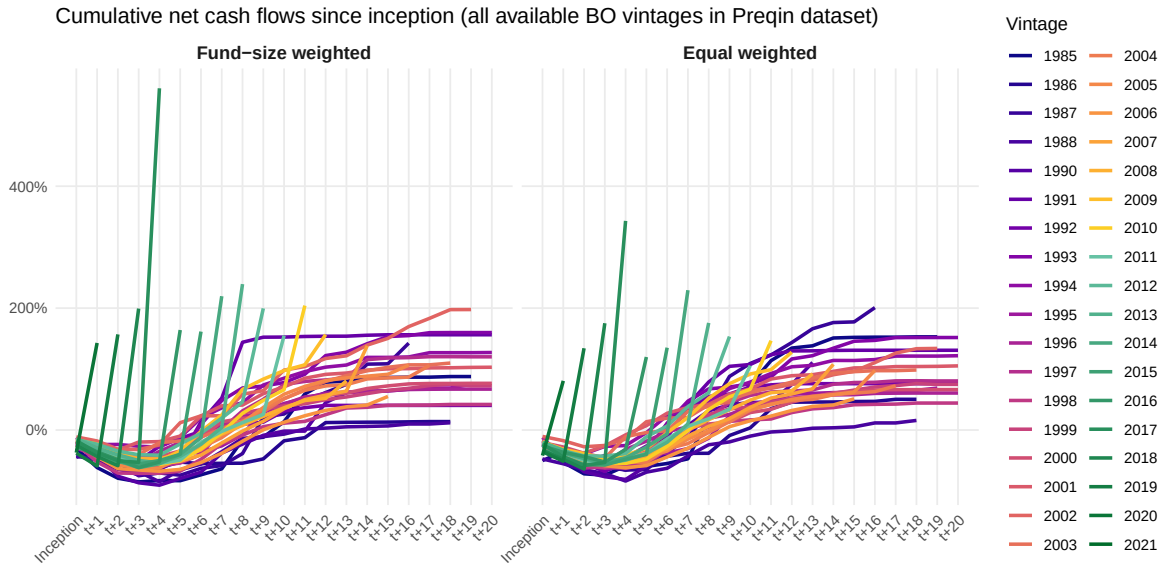

Figure 3: **Cumulative cash flows:** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 3.3, these vintages are excluded from the empirical estimates in Section 4 to limit reliance on unrealized NAVs.

(5), 1988 (6), 1990 (2), 1991 (3), 1992 (6), 1993 (4), 1994 (14), 1995 (11), 1996 (13), 1997 (14), 1998 (29), 1999 (24), 2000 (34), 2001 (19), 2002 (13), 2003 (19), 2004 (24), 2005 (43), 2006 (48), 2007 (44), 2008 (43), 2009 (19), 2010 (31), 2011 (24), 2012 (36), 2013 (34), 2014 (44), 2015 (34), 2016 (54), 2017 (35), 2018 (43), 2019 (63), 2020 (32), 2021 (22). The region distribution is as follows: North America (461), US (433). The strategy distribution is as follows: Buyout (894). The fund currency distribution is as follows: USD (894).

Venture Capital sample. The Venture Capital (North America) sample contains 931 distinct funds spreading over 38 vintage years. The vintage year distribution is as follows: 1982 (1), 1985 (3), 1986 (4), 1987 (2), 1988 (1), 1989 (1), 1990 (4), 1991 (1), 1992 (7), 1993 (6), 1994 (6), 1995 (8), 1996 (14), 1997 (15), 1998 (28), 1999 (44), 2000 (68), 2001 (45), 2002 (26), 2003 (18), 2004 (31), 2005 (34), 2006 (43), 2007 (46), 2008 (36), 2009 (18), 2010 (18), 2011 (22), 2012 (20), 2013 (32), 2014 (31), 2015 (37), 2016 (34), 2017 (48), 2018 (46), 2019 (59), 2020 (44), 2021 (30). The region distribution is as follows: North America (267), US (664). The strategy distribution is as follows: Early Stage (268), Early Stage: Seed (44), Early Stage: Start-up (23), Expansion / Late Stage (123), Venture (General) (451), Venture Debt (22). The fund currency distribution is as follows: USD (931).

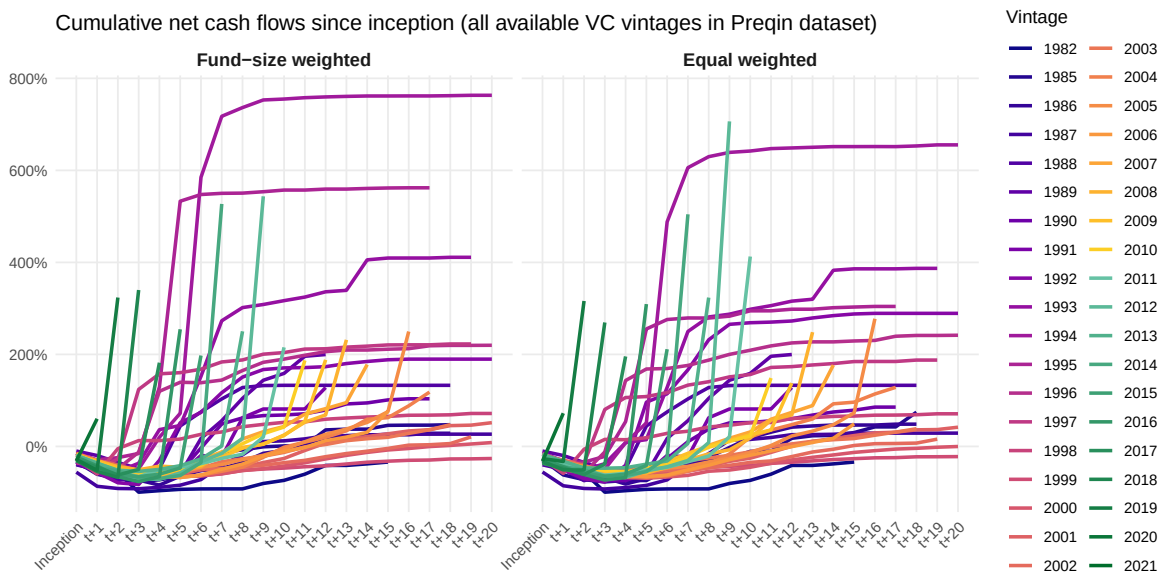

Figure 4: **Cumulative cash flows:** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 3.3, these vintages are excluded from the empirical estimates in Section 4 to limit reliance on unrealized NAVs.

3.2 SDF models and public data

We test a simple linear SDF model similar to Driessen et al. (2012)

$$\Psi^{\mathrm{SL}}_{\tau,t}(\theta) = \frac{\prod_{h=1}^{\tau}\left(1+\alpha+r_{f,h}+\sum_m \beta_m F_{m,h}\right)}{\prod_{h=1}^{t}\left(1+\alpha+r_{f,h}+\sum_m \beta_m F_{m,h}\right)} \tag{19}$$

and an exponential affine SDF model adapted from Korteweg and Nagel (2016)

$$\Psi^{\mathrm{EA}}_{\tau,t}(\theta) = \exp\left[-\sum_{h=\tau+1}^{t}\left(\alpha+\log(1+r_{f,h})+\sum_{m\in M}\beta_m\cdot\log(1+F_{m,h})\right)\right] \tag{20}$$

with monthly (arithmetic) risk-free return r_f, monthly (arithmetic) zero-net-investment portfolio returns F_m, and parameter vector $\theta = (\alpha, \beta)$. Throughout both the simulation and the empirical application, all factor returns and fund cash flows are observed at monthly frequency.

The public market factors that enter our SDF draw on the US dataset of the recently popularized q^5 investment factor model sourced from `http://global-q.org/factors.html` (Hou et al., 2015, 2021). According to Hou et al. (2019) this family of new "investment" factors subsumes the more popular Fama-French factors (Fama and French, 2015, 2018), which indicates higher explanatory power of the Hou et al. (2021) model. The q^5 five-factor model includes the market excess return (MKT), a size factor (ME), an investment factor (IA), a return on equity factor (ROE), and an expected growth factor (EG). The maximum date in the q^5 investment factor dataset is 2024-12-31.

To avoid overfitting, we restrict our analysis to six simple SDF models that contain {MKT} alone or {MKT} plus {ME or IA or ROE or EG or Alpha}. In Equation (13), we use the quadratic loss function $L(x) = x^2$.

In our optimization procedure, we limit the alpha intercept α to the range $[-2\%, +2\%]$ per month to avoid the "exploding alpha" issue documented in (Driessen et al., 2012, Footnote 8): when α is unrestricted, an arbitrarily large value can almost perfectly price any cash flow stream (except its first component), rendering the beta risk factors irrelevant.

3.3 The impact of unseasoned funds and reported NAVs

Before defining our baseline sample, we emphasize a structural limitation of including recent vintages: treating the final reported NAV as a terminal cash flow is problematic not only because of potential staleness, but also because it represents an expected continuation value rather than a realized payoff. This formally violates the actual-divestment requirement of Assumption 1, meaning that whenever NAV is used as a terminal cash flow, the estimator technically targets an approximation. We alleviate this issue by implementing a strict maximum vintage-year cutoff to ensure the sample primarily consists of seasoned funds with minimal remaining NAVs.

To illustrate this mechanism, Figure 5 presents a vintage-year cutoff sensitivity analysis separately for BO and VC under fund-value weighting (FW). A clear upward-drift pattern is present in both sub-samples: as more recent, unseasoned vintages are included, the estimated market beta increases monotonically. For BO under value weighting, $\hat{\beta}_{\text{MKT}}$ at the 0-year Horizon (see Remark 1) rises from approximately 1.23 with a 2011 vintage cutoff to roughly 2.42 with the full sample (2021). Notably, most of the increase is concentrated in vintages after 2016, where $\hat{\beta}_{\text{MKT}}$ jumps sharply from 1.53 (2016 cutoff) to 2.34 (2017 cutoff), reflecting the disproportionate influence of young, unrealized funds. For VC, the drift is similarly dramatic: the FW estimate at the 0-year Horizon climbs from approximately 1.57 (2011 cutoff) to 2.33 with the full sample. Discounting all final NAVs by 50% substantially flattens the upward trend: the BO FW estimate range compresses to 1.09–1.65, and the VC FW range to 1.41–1.54, consistent with the view that the drift is partly attributable to the outsized mechanical reliance on subjectively reported Net Asset Values for young funds. These findings reinforce the sample restriction to the 2010 maximum vintage year applied throughout the remainder of this section. Figure 6 confirms that this distortion also affects the two-factor models, where unseasoned post-2016 vintages severely inflate the estimated intercept α. For Buyout funds at short horizons, this upward bias mechanically drives α into its +2% upper parameter boundary, an estimation collapse that is effectively mitigated by the 50% NAV discount.

Remark 1 (Definition of Horizon)**.** *Throughout the empirical section, we use the term* Horizon *H to denote the maximum compounding horizon in years used in the set of discounting dates $\mathcal{T}$ (cf. Equation* (12)*). Formally, starting from the fund inception date t_i^0, estimating a model with a Horizon H implies that the estimator minimizes the average pricing error over the set of discounting dates $\mathcal{T}_{i,H} = \{t_i^0, t_i^0+1, \ldots, \min(t_i^0+12H, T_i^{\max})\}$, formally denoted as $\bar{\epsilon}_i^{(H)}$. Note that the Horizon has as upper bound the max cash flow date per fund or portfolio in our empirical application. This Horizon in years serves as x-axis in all subsequent empirical and simulation charts.*

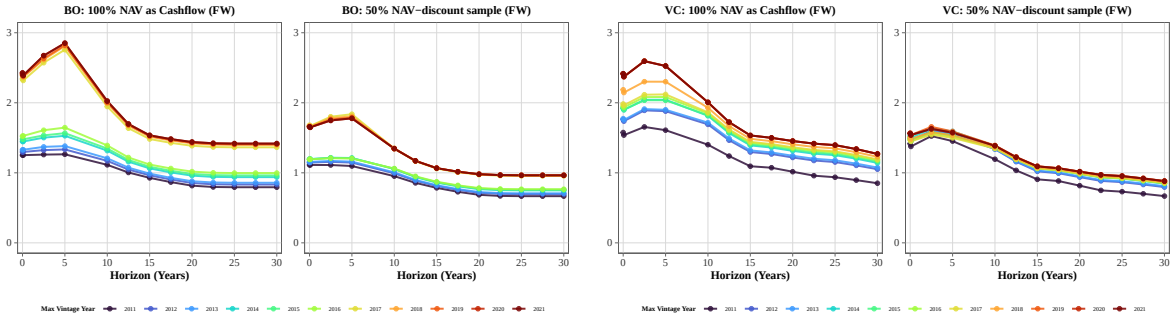

Figure 5: **Sensitivity analysis (MKT factor):** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

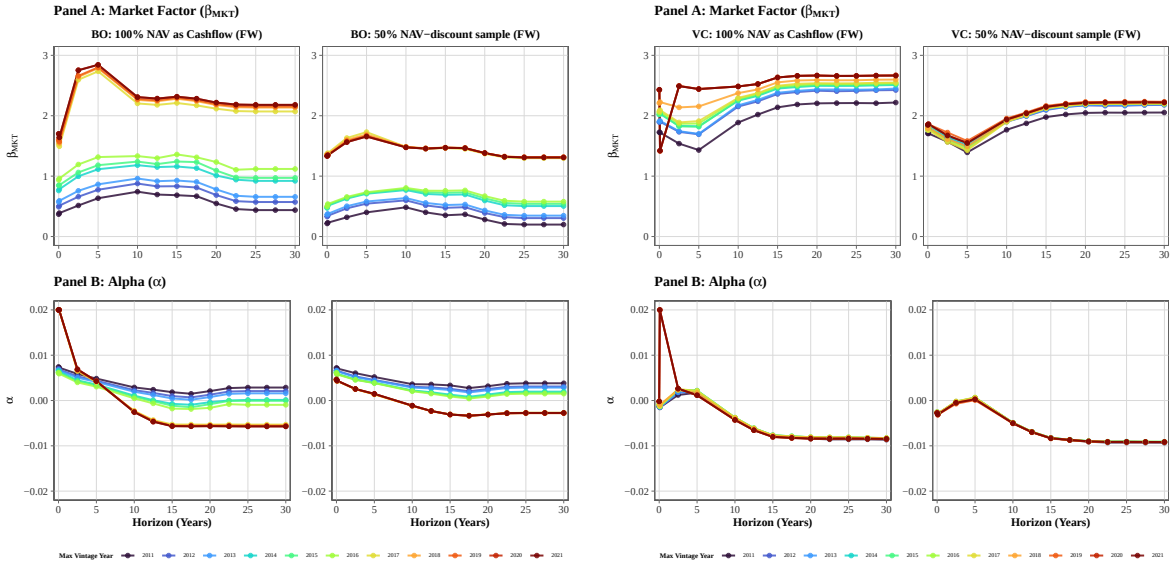

Figure 6: **Sensitivity analysis (α + MKT factor):** Two-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

3.4 NAV discount sensitivity analysis

To further quantify the sensitivity of factor loading estimates to terminal Net Asset Value treatment, we conduct a systematic NAV discount analysis on the seasoned sample (maximum vintage year 2010). Figure 7 reports single-factor MKT estimates under five NAV treatments: the baseline (full reported NAV as final cash flow), 25%, 50%, and 75% constant discounts applied to all final NAVs, and a complete NAV write-off (100% discount).

For BO, the estimated market beta decreases monotonically with the NAV discount. At the 0-year Horizon under EW, $\hat{\beta}_{\text{MKT}}$ drops from 1.26 (no discount) to 1.19 (25% discount), 1.12 (50%), 1.03 (75%), and 0.94 (100% write-off); under FW, the corresponding range is 1.20 to 0.98. At the 15-year Horizon, the EW estimate ranges from 0.81 (no discount) to 0.55 (100% write-off), while the FW estimate spans 0.84 to 0.62. The qualitative conclusion is robust across all NAV treatments: the single-factor market model remains economically plausible and comparatively stable.

For VC under EW, the pattern is analogous but the sensitivity is more muted in percentage terms due to the higher baseline beta: the 0-year estimate declines from 1.88 (no discount) to 1.84 (25%), 1.80 (50%), 1.77 (75%), and 1.74 (100%); at the 15-year Horizon, from 1.30 (no discount) to 1.09 (100% write-off). Under FW, the sensitivity is more pronounced, particularly at longer Horizons: the 0-year estimate ranges from 1.59 to 1.34, while the 15-year estimate declines from 1.07 to 0.73.

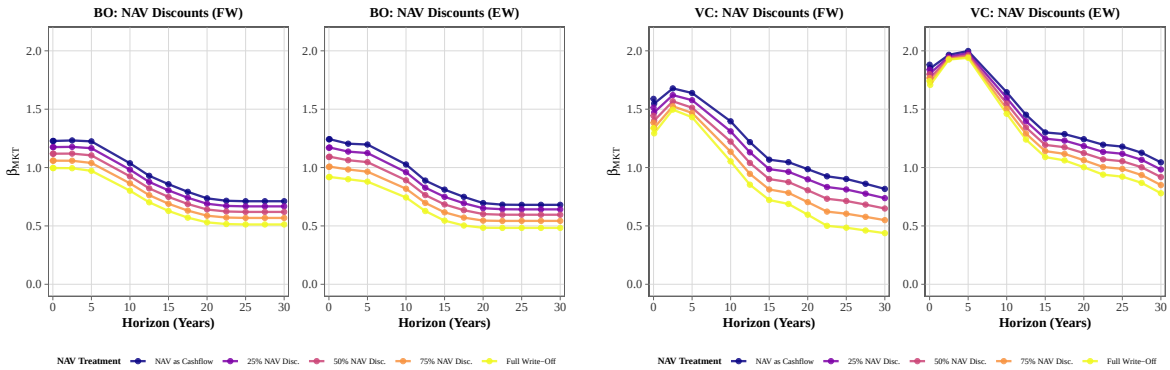

Figure 7: **NAV discount sensitivity (*MKT* factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off.

This analysis demonstrates that terminal NAV treatment has a material but bounded effect on estimated factor loadings. The monotonic relationship is consistent with the intuition that reported NAVs mechanically inflate estimated betas: funds whose final cash flow is a subjective valuation rather than a realized exit price appear more systematically risky because their NAVs partially co-move with public market indices. Importantly, even under the extreme assumption of a complete NAV write-off, the first qualitative conclusions of this subsection can be summarized as follows: BO maintains a market beta near or below unity, while VC retains a substantially higher systematic exposure. We do not impose a model-based residual-value adjustment, as

this would constitute a separate research agenda (Jenkinson et al., 2020; Brown et al., 2023). Instead, the range spanned by the five NAV treatments provides a transparent envelope of estimation uncertainty attributable to terminal value measurement.

3.5 Simulation study

Our Monte Carlo experiments examine the finite-sample behavior of our estimation methodology through bias and variance.[2] Specifically, we investigate five key dimensions of the estimation framework: (i) the finite-sample efficiency gains of forming vintage-year portfolios versus using individual funds; (ii) the tradeoff between expanding the time-series dimension (number of vintage years) and the cross-sectional dimension (funds per vintage); (iii) the robustness of the estimates to the functional specification of the SDF (simple linear vs. exponential affine); (iv) the behavior of multivariate identification under idiosyncratic noise using two-factor models; and (v) as the central design choice in the paper, the optimal choice of the maximum compounding Horizon (i.e., the set of discounting dates $\mathcal{T}$).

We begin from a baseline simulation that uses historical q-investment factor paths from 1986 to 2005 and simulates 20 funds for each of these 20 vintage years. Thus, the baseline conditions on a realized 1986–2005 factor path rather than simulating market factors. The subsequent paragraphs compare this baseline to alternative scenario blocks that vary one margin at a time, such as the number of funds per vintage, the vintage span, the factor specification, or the fund lifetime; the full scenario definitions are listed in the simulation configuration file in the online repository. In particular, the "extended vintage" scenarios use vintages 1967–2005, so $V = 39$, whereas the baseline uses vintages 1986–2005, so $V = 20$. Each fund contains 15 deals with equal investment amounts and exactly one divestment cash flow. Deals are entered within the first five years of the fund lifetime following a discrete uniform distribution and afterward held between one to ten years again uniformly distributed. The deal returns are generated by the simple linear or exponential affine SDF models described in Equations (19) and (20). In the base case, we just use the MKT factor with $\beta_{\text{MKT}} = 1$ and in each month, add a normal i.i.d. error term with monthly standard deviation $\sigma = 0.2$ (annualized variance $12 \cdot 0.2^2 = 0.48$, corresponding to an annualized volatility of $0.2\sqrt{12} \approx 69\%$) and zero mean. In the exponential affine case, we adjust the lognormally distributed error mean to zero by subtracting $0.5\sigma^2$. If a negative return exceeds -100%, the company defaults with a zero exit cash flow. In contrast, the error term in the simulations of Driessen et al. (2012) is more well-behaved as it follows a shifted lognormal distribution that, even with arbitrarily high error term variance, just allows for returns below say -99%, if the market return is close to its lower bound (see equation 9 in their online appendix). We specifically analyze the remaining divergence from their simulation design in Appendix B.1. In our simulation scenarios, we evaluate the estimator across a full grid of compounding Horizons, where the maximum month in the set of discounting dates $\mathcal{T}$ ranges from 0 up to 30 years (0-360 months). To assess our estimator's bias and variance, we simulate 1000 test scenarios for both vintage-year and single-fund portfolios.

[2] As with any simulation study, it investigates the ability to identify the assumed data generating process rather than validating the corresponding SDF model itself.

Cross-sectional unit i: As anticipated in Subsection 2.3, vintage year portfolios demonstrate lower bias and variance than individual funds. Under the simple linear SDF and a 15-year Horizon (maximum month 180), the mean (standard deviation) of the coefficient estimate $\hat{\beta}_{\text{MKT}}$ is 0.966 (0.188) for vintage year portfolios, compared to 0.887 (0.359) for individual funds. The superiority of the portfolio approach is especially striking when evaluating the entire Horizon curve in Figure 8. Estimates for individual funds are highly unstable, exhibiting severe upward bias at short Horizons ($\hat{\beta}_{\text{MKT}} \approx 1.5$) and deteriorating to negative values at long Horizons. In contrast, vintage year portfolios yield substantially more stable estimates that remain within a narrow, economically plausible range around the true value of 1.0. The portfolio approach consistently achieves a lower standard deviation across all simulated Horizons, confirming that grouping funds effectively mitigates idiosyncratic noise.

This trade-off between bias/variance reduction and sample size has important implications. While forming vintage year portfolios substantially improves the accuracy of the estimates, it dramatically reduces the number of cross-sectional units, thereby weakening the relevance of asymptotic approximations. This constraint likely explains the choice of Korteweg and Nagel (2016) to rely on individual funds as cross-sectional units in their asymptotic SHAC framework, enabling them to exploit a larger sample size for smaller standard errors.

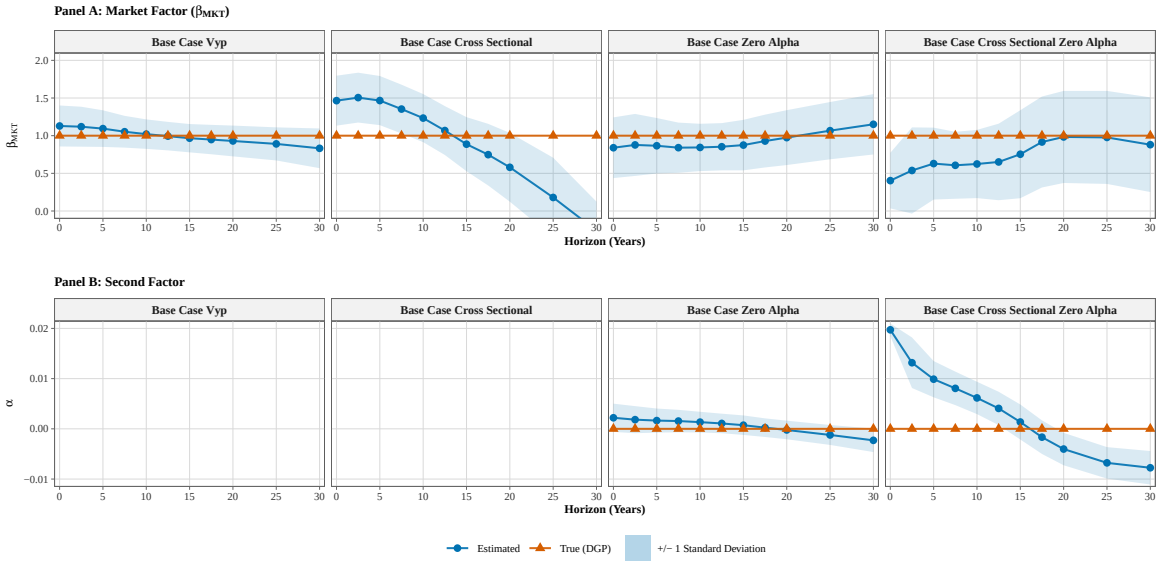

Figure 8: **Simulation (β_{MKT}, α):** Single-fund vs. vintage-year portfolios.

Varying vintages V and portfolio sizes n/V: A fundamental question in private equity panel data econometrics is whether precision is best improved by extending the time-series dimension (V, number of vintage years) or the cross-sectional dimension (n/V, number of funds per vintage portfolio). For this comparison block, we intentionally switch from the overall 20-fund baseline to a dedicated reference scenario with larger portfolios, so that changes in V and n/V can be separated more cleanly. To assess this tradeoff, we compare five data configurations (depicted in Figures 9 and 10): (1) a reference scenario with $V = 20$, $n/V = 40$; (2) an extended vintage

sample with small portfolios ($V = 39$, $n/V = 10$); (3) an extended vintage sample with medium portfolios ($V = 39$, $n/V = 20$); (4) an early-vintage sub-sample ($V = 10$, years 1986–1995, $n/V = 20$); and (5) a late-vintage sub-sample ($V = 10$, years 1996–2005, $n/V = 20$).

In our simulation design, increasing the number of cross-sectional units per portfolio (n/V) is substantially more effective at suppressing estimator variance than adding more vintage years (V). For the single-factor model at the 15-year Horizon, transitioning from 10 funds to 20 funds per vintage over a 39-year period (scenarios 2 vs. 3) reduces the standard deviation of $\hat{\beta}_{\text{MKT}}$ by roughly 22%, from 0.218 to 0.169. In contrast, shrinking the time-series dimension from 39 years to just 10 years (while holding $n/V = 20$ constant) elevates the variance only moderately for the early-vintage sub-sample ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.194$ for 1986–1995), though the late-vintage sub-sample (1996–2005) exhibits considerably higher variance ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.432$), reflecting the sensitivity to the realized macro environment. This hierarchy of variance reduction is even more pronounced in the two-factor setting, where idiosyncratic cash flow noise severely impairs multivariate identification: increasing n/V aggressively dampens the spurious α variance. Consequently, our findings provide an empirical rationale for the finite-sample superiority of the Driessen et al. (2012) infill asymptotics (fixing V, letting $n_v \to \infty$) over approaches that rely purely on time-series aggregation.

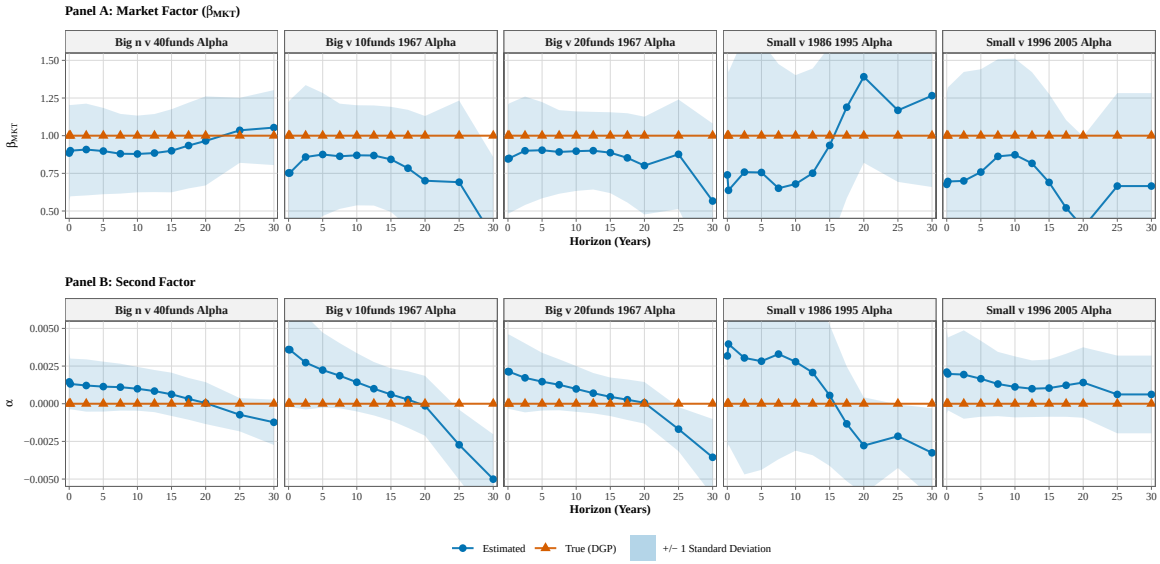

Figure 9: **Simulation (β_{MKT}, α):** Varying vintages V and portfolio sizes n/V.

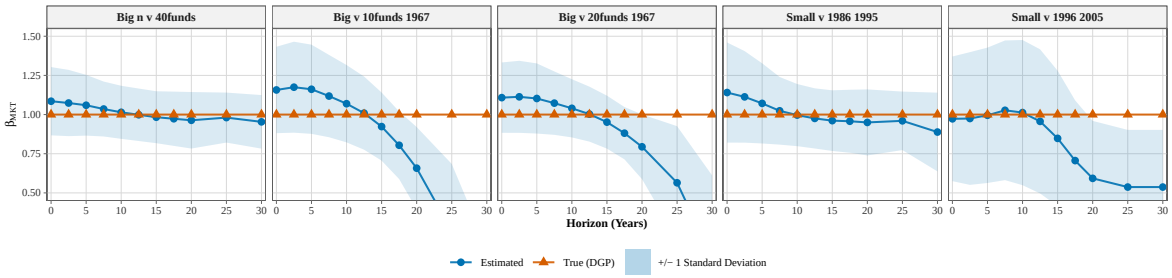

Figure 10: **Simulation ($\beta_{\mathbf{MKT}}$):** Varying vintages V and portfolio sizes n/V.

SDF model Ψ: In Figure 11, we compare the simple linear SDF model against the exponential affine specification. For the single-factor model, the bias is minimized at the 12.5-year Horizon (150 months), where both functional forms perform almost identically and yield estimates extremely close to the true value of 1.0: the exponential affine yields a mean market beta of 0.999 (0.182), matching the simple linear's 0.994 (0.189). For the two-factor (MKT+Alpha) model at this same 12.5-year Horizon, both functional forms again yield nearly identical parameter results, though with an attenuated market beta ($\hat{\beta}_{\text{MKT}} \approx 0.85$) and a minor spurious positive alpha ($\hat{\alpha} \approx 0.001$). In the two-factor case, the market beta bias is actually minimized later, around the 20-year Horizon ($\hat{\beta}_{\text{MKT}} \approx 0.97$, $\hat{\alpha} \approx -0.0002$).

However, a salient difference emerges at extremely short compounding Horizons. At a 1-month Horizon (resembling an NPV-only discounting framework), the exponential affine formulation deteriorates much more severely in both models, manifesting a larger upward bias (e.g., in the single-factor case, $\hat{\beta}_{\text{MKT}} = 1.21$ vs. 1.13) and elevated variance (0.39 vs. 0.27) compared to the linear model. Overall, the simulation evidence does not indicate a systematic finite-sample advantage of exponential affine SDFs (Gourieroux and Monfort, 2007; Bertholon et al., 2008) in our framework. If anything, the simple linear model appears slightly better behaved in the reported designs.

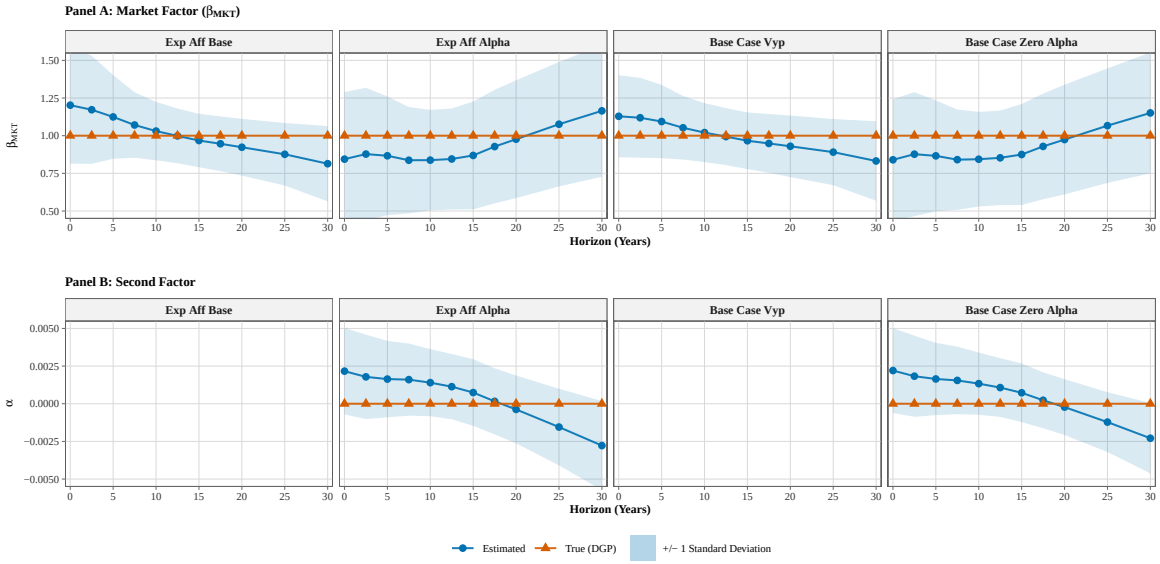

Figure 11: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Exponential affine vs. linear models.

Two-factor models (q^5 factors): Figure 12 reports simulation results for two-factor models that pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG), with true loadings $\beta_{\text{MKT}} = 1$ and $\beta_{\text{2nd}} = 0.5$. A common pattern emerges across all four specifications: the second-factor loading is substantially overestimated at short Horizons and attenuated toward (or beyond) the true value at long Horizons, while the market factor absorbs a compensating bias in the opposite direction. Estimation variance is markedly larger than in the single-factor case, reflecting the collinearity between correlated factors in a limited cross-section of vintage year portfolios. Table 4 summarizes the key estimates at the shortest Horizon and at 15 years, which is approximately the MSE-minimizing Horizon for most models; we highlight the most notable factor-specific features below.

The ME model is the only specification in which $\hat{\beta}_{\text{2nd}}$ undergoes a sign change (at roughly 17.5 years), declining from 0.86 to −0.02 over the full Horizon range, a severe attenuation that would qualitatively alter economic conclusions. The IA model exhibits a large second-factor bias at short Horizons ($\hat{\beta}_{\text{IA}} \approx 1.06$ vs. truth 0.5) but both loadings converge close to their true values near the 17.5-year Horizon ($\hat{\beta}_{\text{MKT}} = 0.97$, $\hat{\beta}_{\text{IA}} = 0.52$). The EG model exhibits the largest overall estimation variance ($\sigma(\hat{\beta}_{\text{EG}}) \approx 1.5$ at the shortest Horizon) but achieves a reasonably accurate joint estimate around 15–17.5 years ($\hat{\beta}_{\text{MKT}} = 0.91$–$1.02$, $\hat{\beta}_{\text{EG}} = 0.47$–$0.56$). The ROE model is the exception: both loadings remain persistently biased at all Horizons, with $\hat{\beta}_{\text{MKT}}$ never exceeding 0.91 and $\hat{\beta}_{\text{ROE}}$ never falling below 0.62. These results suggest that empirical researchers should exercise caution when interpreting two-factor SDF estimates for private equity, and that the MSE-minimizing Horizon is model-dependent, typically falling in the 12.5–17.5-year range.

	$\hat{\beta}_{\text{MKT}}$		$\hat{\beta}_{\text{2nd}}$		$\sigma(\hat{\beta}_{\text{MKT}})$		$\sigma(\hat{\beta}_{\text{2nd}})$	
Model	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr
ME + MKT	1.15	0.97	0.87	0.59	0.28	0.21	0.46	0.40
IA + MKT	0.97	0.92	1.06	0.64	0.34	0.30	0.74	0.53
ROE + MKT	0.88	0.83	0.98	0.69	0.43	0.49	0.61	0.49
EG + MKT	0.87	0.91	0.99	0.56	0.61	0.45	1.53	0.31

True values: $\beta_{\text{MKT}} = 1$, $\beta_{\text{2nd}} = 0.5$.

Table 4: Two-factor simulation summary: mean estimates and standard deviations at the shortest (0 yr) and the 15-year Horizon.

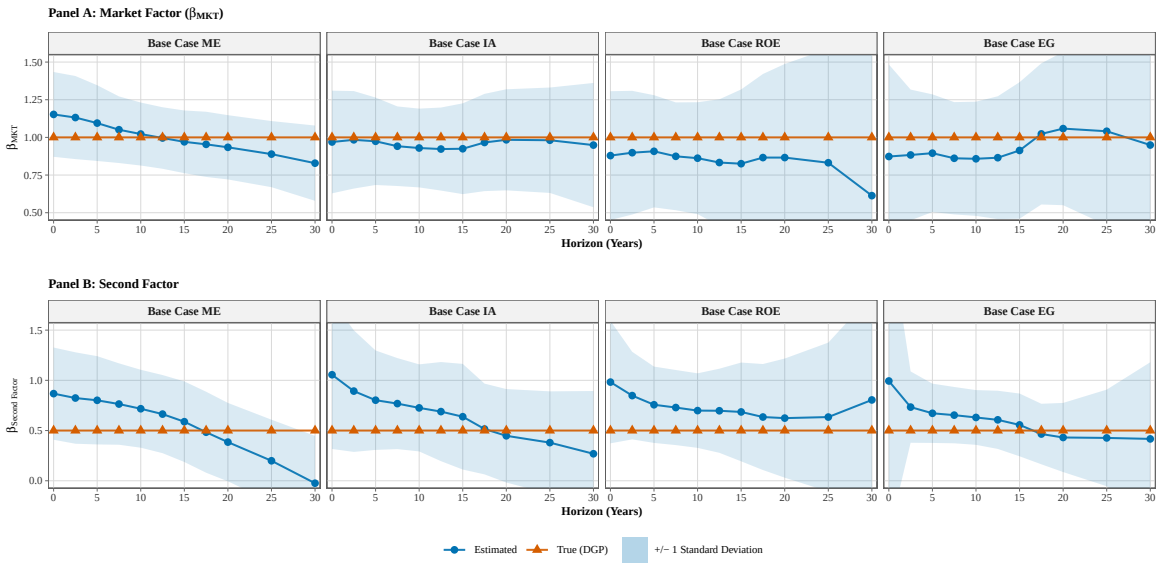

Figure 12: **Simulation (q^5 factors):** Two-factor models using VYP.

Varying fund lifetimes: Figure 13 illustrates how the optimal Horizon shifts in response to abbreviated fund lifetimes. As anticipated, the intercept of the alpha parameter decreases for shorter fund lifetimes relative to our baseline scenario of a five-year investment period and a ten-year maximum holding period. Specifically, for the shortest examined fund lifetime of two years (comprising a one-year investment period and a one-year maximum holding period), the beta bias is minimized at a 12.5-year Horizon, while the alpha bias is minimized at a 5-year Horizon. Similarly, for a fund with a two-year investment period and a two-year maximum holding period, the minimum beta bias occurs at a 15-year Horizon, and the minimum alpha bias at a 7.5-year Horizon. Consequently, these results suggest that for funds with short lifespans, the Horizon that minimizes bias can extend significantly beyond the fund's actual lifetime.

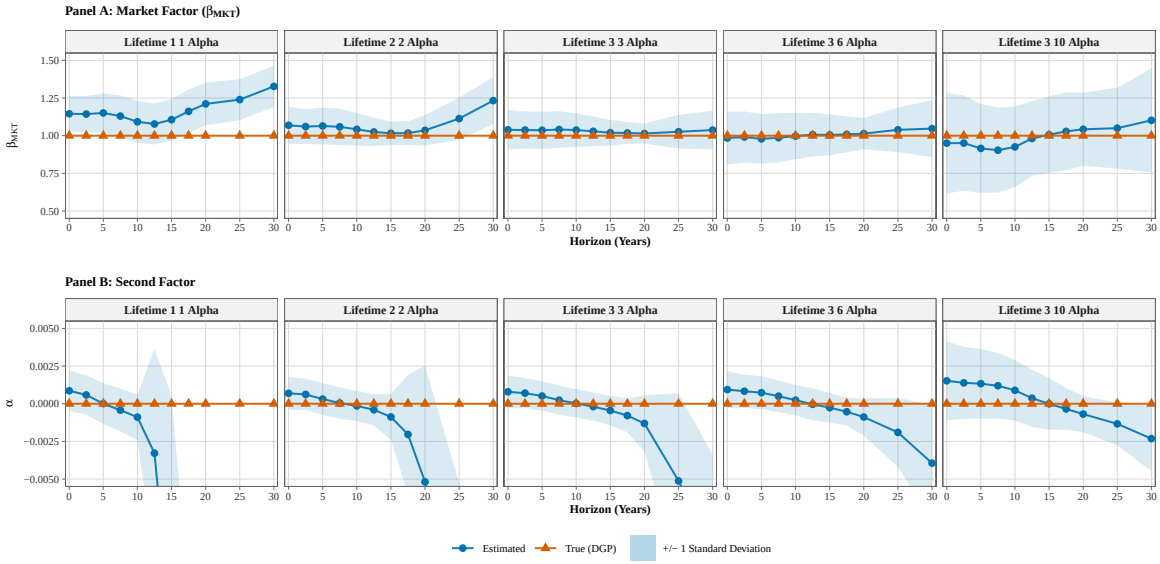

Figure 13: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying fund lifetimes using VYP with the following yearly investment-period/holding-period pairs: 1/1, 2/2, 3/3, 3/6, and 3/10 (from left to right).

Size of set $\mathcal{T}$ (Maximum Compounding Horizon): A central simulation finding concerns the choice of the maximum compounding Horizon, i.e., the size of the set of discounting dates $\mathcal{T}$. From a purely theoretical perspective (cf. Section 2.1), discounting all cash flows exclusively to the fund inception date, corresponding to a zero-month or NPV-only Horizon in our notation, should yield unconditionally unbiased estimates under the deal-level exact-pricing restriction (Equation (1)). This minimal choice for $\mathcal{T}$ corresponds exactly to the established Driessen et al. (2012) framework.

However, our simulation results highlight a tension between structural fidelity and finite-sample recoverability: extending the set of discounting dates into the fund lifecycle can materially stabilize the estimator in the reported designs. Across most model specifications (see Figures 12 and 11), bias is minimized not at the theoretical 0-year Horizon, but rather around a 12.5 to 15-year compounding Horizon. At extremely short Horizons, idiosyncratic cash flow noise interacts non-linearly with the estimation procedure, inducing severe bias in factor loadings. Averaging pricing errors over multiple future dates therefore acts as a finite-sample stabilization device, while also introducing the timing component discussed in Section 2.1.

In the reported designs, this stabilization improves the mean estimate far more than it reduces standard errors. While estimator variance traces a shallow U-shape with a minimum near the 15-year Horizon (see Table 4) for fund with a lifetime of 15 years, this reduction is secondary to the striking stabilization of the parameter bias. However, our simulations for shorter fund lifetimes indicate that determining the optimal horizon requires tailored simulation designs rather than a simple heuristic assuming the optimal horizon always coincides with the fund's life. We further analyze these puzzling results from a theoretical perspective in Appendix A, detailing the finite-sample stabilization under $\mathcal{T}$-averaging and the role of the Compounding-Horizon

Wedge.

Simulation conclusion: To conclude, our simulation study supports several implementation choices used in the empirical application. First, vintage-year portfolio formation helps improve estimator precision relative to single-fund estimation. Second, in the reported designs, increasing the number of funds per vintage appears more effective in controlling estimator variance than increasing the number of vintages (for i.i.d. error terms). Third, the simulation evidence does not indicate a clear finite-sample advantage of the exponential affine SDF over the simple linear SDF in this multi-period framework. Fourth, averaging pricing errors over multiple dates materially reduces the short-Horizon bias of the inception-only criterion in the reported designs, even though Appendix A shows that this stabilization comes with a horizon-specific target shift.

Accordingly, in the empirical application we use vintage-year portfolios, the simple linear SDF, emphasize one-factor MKT specifications, treat two-factor results as exploratory, and focus on Horizons around 12.5–15 years while also reporting sensitivity over a broader grid.

3.6 Role of asymptotic inference

To assess the parameter significance in the empirical application, we compute the asymptotic standard errors as outlined in Subsection 2.3. For the Bartlett kernel bandwidth b_V, we set $b_V = 12$ years, so pairs of funds with absolute vintage-year differences exceeding 12 receive zero kernel weight in the SHAC covariance estimator. The asymptotic results provide the benchmark large-sample approximation.

At the same time, private-equity samples remain small in the vintage-year and cross-sectional dimensions, and model choice is often data-driven. For that reason, the empirical analysis complements asymptotic inference with out-of-sample validation and stability diagnostics. To account for the dependency between funds from adjacent vintage years caused by overlapping fund cash flows, we draw on hv-block cross-validation (Racine, 2000). As Racine (2000) demonstrates, hv-block cross-validation is a consistent model-selection criterion for dependent data: the h-block gap between training and validation sets ensures approximate independence, while the v-block provides a sufficiently large validation sample within each fold. In our implementation, each cross-validation fold partitions the data into three disjoint groups: (i) a *validation set* (v-block) comprising funds from three consecutive vintage years (e.g., 2000, 2001, 2002), on which the out-of-sample pricing error is evaluated; (ii) a *buffer zone* (h-block) of three vintage years on each side of the v-block (e.g., 1997–1999 and 2003–2005), which is discarded entirely to break the dependence between training and validation data caused by overlapping fund cash flows; and (iii) an *estimation set* consisting of all remaining vintage years (e.g., 1985–1996 and 2006–end), on which the model parameters are estimated. This three-way split is repeated across rolling vintage-year windows, as detailed in Table 5. The number of active folds adapts dynamically to the selected maximum vintage year cutoff in any given estimation run. For instance, when estimating the full sample through 2021, we apply ten folds; when estimating a heavily seasoned sample truncated at 2010 to avoid reliance on subjective interim NAVs, the framework naturally truncates to seven active folds.

Remark 2 (Cross-validation as stability diagnostic)**.** *The hv-block cross-validation quantities reported throughout this paper serve as* ***stability diagnostics and model-selection criteria*** *rather than inferential standard errors. They summarize the fold-to-fold variation in parameter estimates under dependence-reducing sample splits and are informative about model sensitivity and effective parameter curvature, but they do not provide a valid null distribution for the full-sample estimator. For formal statistical inference (standard errors and hypothesis tests), we rely on the SHAC-based asymptotic benchmark of Section 2.3. In practice, hv-block cross-validation complements the asymptotic analysis by (i) diagnosing model stability and out-of-sample performance without relying on asymptotics and (ii) flagging over-parameterized specifications that appear well-identified in-sample but collapse out-of-sample.*

training.before	h-block.before	v-block	h-block.after	training.after
estimation	remove	validation	remove	estimation
start-1984	1985,1986,1987	1988,1989,1990	1991,1992,1993	1994-end
start-1987	1988,1989,1990	1991,1992,1993	1994,1995,1996	1997-end
start-1990	1991,1992,1993	1994,1995,1996	1997,1998,1999	2000-end
start-1993	1994,1995,1996	1997,1998,1999	2000,2001,2002	2003-end
start-1996	1997,1998,1999	2000,2001,2002	2003,2004,2005	2006-end
start-1999	2000,2001,2002	2003,2004,2005	2006,2007,2008	2009-end
start-2002	2003,2004,2005	2006,2007,2008	2009,2010,2011	2012-end
Inactive in empirical pre-2011 sample:				
start-2005	2006,2007,2008	2009,2010,2011	2012,2013,2014	2015-end
start-2008	2009,2010,2011	2012,2013,2014	2015,2016,2017	2018-end
start-2011	2012,2013,2014	2015,2016,2017	2018,2019,2020	2021-end

Table 5: Partitions used for hv-block cross-validation. The number of active folds truncates dynamically based on the maximum vintage year cutoff of the specific estimation run.

4 Empirical results for BO and VC data

This section presents the empirical SDF estimates for Buyout (BO) and Venture Capital (VC) funds and uses them to assess which specifications are stably identified in the current cash-flow panel. As the simulation study in Subsection 3.5 confirms, we use vintage-year portfolios (VYPs) to estimate simple linear SDF models for the BO and VC sub-samples individually. We restrict the data to North American, USD-denominated funds from the Preqin dataset described in Subsection 3.1. This restriction ensures consistency between fund cash flows and the U.S. q-investment factors used throughout the analysis. Moreover, we only consider funds with a maximum vintage year of 2010, to reduce the reliance on unrealized NAVs (see Section 3.3).

We analyze BO and VC as two distinct asset classes with fundamentally different risk profiles and report both equal-weighted (EW) and value-weighted (FW) estimates across a

grid of compounding Horizons. Our baseline specification is the single-factor MKT model estimated on vintage-year portfolios. Two-factor models that add an alpha intercept or a single q-investment factor serve as robustness and exploratory extensions.

For each model specification, we report asymptotic standard errors (SE_A) from the SHAC framework alongside hv-block cross-validation fold dispersions (SD_{CV}), defined as the standard deviation of parameter estimates across hold-out folds. Throughout the empirical analysis, t_A denotes the t-ratio based on asymptotic standard errors; SR_{CV} denotes the signed stability ratio $\mathrm{SR}_{CV} := \hat{\theta}/\mathrm{SD}_{CV}$, where $\hat{\theta}$ is the corresponding full-sample estimate. Because SR_{CV} preserves the sign of $\hat{\theta}$, negative values simply indicate negative estimated coefficients; its absolute value $|\mathrm{SR}_{CV}|$ measures the coefficient magnitude relative to fold-to-fold dispersion. We use SR_{CV} descriptively rather than as a formal test statistic. As discussed in Remark 2, SR_{CV} complements asymptotic inference as a stability diagnostic. We report asymptotic significance markers for completeness, but when SE_A is erratic, boundary solutions arise, or the cross-validation diagnostics indicate instability, we do not treat isolated significant t_A values as decisive.

All R code and data (including the previous simulations) are available in an online repository: `https://github.com/quant-unit/Fundwise_SDF/tree/master/r_project`.

4.1 Baseline: single market-factor models

Tables 6–7 and Figure 14 report the single-factor MKT model for BO and VC across Horizons from 0 to 30 years. As anticipated by the simulation study, the market beta declines with the compounding Horizon for both asset classes.

For BO under EW, $\hat{\beta}_{\mathrm{MKT}}$ ranges from approximately 1.20–1.24 at short Horizons (0–5 years) and decreases to 0.68 at 25–30 years; under FW, the corresponding range is 1.22–1.23 down to 0.71. The divergence between asymptotic and cross-validation diagnostics persists: asymptotic t-ratios (t_A) are moderate at short Horizons (e.g., 0.94–1.05 under EW at 0–5 years) and decline further at longer Horizons, while the cross-validation stability ratios remain high throughout (SR_{CV} ranging from 4.98 to 10.01 under EW and from 2.59 to 8.09 under FW), indicating comparatively stable estimates across data subsets.

For VC, the market beta is substantially higher across all Horizons, consistent with venture capital's greater sensitivity to systematic risk. Under EW, $\hat{\beta}_{\mathrm{MKT}}$ reaches approximately 1.97–2.00 at 2.5–5 years and converges toward 1.05–1.18 beyond 25 years; under FW, the corresponding range is 1.66 down to 0.82–0.90. Despite the higher parameter variability reflected in larger asymptotic standard errors (SE_A of 1.28–1.83), the cross-validation stability diagnostics indicate comparatively stable estimates (SR_{CV} of 3.65–4.55 under EW; 3.53–7.57 under FW), although somewhat lower than the BO stability ratios due to the more heterogeneous VC fund universe. As horizon-specific diagnostics rather than direct structural estimates, the converged long-horizon coefficients (BO $\approx$ 0.68, VC $\approx$ 1.1 under EW) remain economically interpretable: BO funds, which invest in mature companies with stable cash flows, load less on market risk than VC funds, which back early-stage ventures with highly uncertain outcomes.

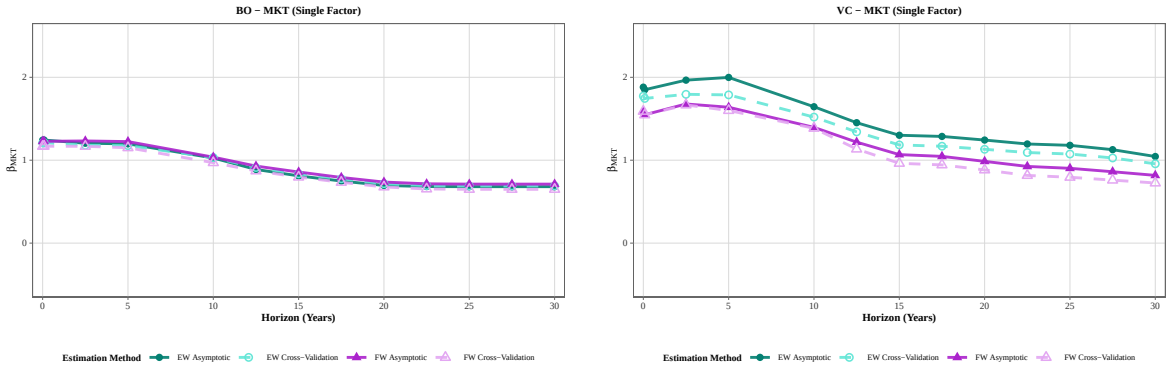

Figure 14: **Empirical (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

Table 6: Empirical Estimates: BO — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.242	1.322	0.121	0.94	10.01	1.228	1.163	0.144	1.06	8.09
2.5 yr	1.204	1.152	0.131	1.05	9.00	1.232	1.165	0.174	1.06	6.72
5 yr	1.197	1.146	0.147	1.04	8.01	1.224	1.161	0.199	1.05	5.78
10 yr	1.027	1.618	0.190	0.63	5.43	1.037	1.790	0.234	0.58	4.15
12.5 yr	0.888	1.655	0.227	0.54	4.03	0.929	1.609	0.254	0.58	3.43
15 yr	0.811	1.514	0.227	0.54	3.68	0.857	1.788	0.254	0.48	3.14
17.5 yr	0.749	1.413	0.213	0.53	3.64	0.792	1.690	0.257	0.47	2.86
20 yr	0.696	2.044	0.173	0.34	4.14	0.736	1.738	0.255	0.42	2.66
25 yr	0.681	1.992	0.139	0.34	4.98	0.712	1.682	0.250	0.42	2.59
30 yr	0.681	1.992	0.139	0.34	4.98	0.712	1.682	0.250	0.42	2.59

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|\mathrm{SR}_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

Table 7: Empirical Estimates: VC — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.881	1.771	0.393	1.06	4.51	1.588	1.657	0.352	0.96	4.51
2.5 yr	1.966	1.809	0.446	1.09	4.02	1.678	1.657	0.452	1.01	3.69
5 yr	1.998	1.829	0.488	1.09	3.67	1.639	1.655	0.434	0.99	3.68
10 yr	1.645	1.280	0.358	1.28	4.24	1.396	1.948	0.315	0.72	4.39
12.5 yr	1.452	1.279	0.299	1.14	4.49	1.218	1.968	0.150	0.62	7.57
15 yr	1.301	1.356	0.260	0.96	4.55	1.068	6.471	0.138	0.16	6.98
17.5 yr	1.287	1.349	0.275	0.95	4.25	1.046	5.758	0.148	0.18	6.35
20 yr	1.243	1.328	0.278	0.94	4.07	0.986	4.384	0.160	0.22	5.52
25 yr	1.179	1.300	0.267	0.91	4.03	0.902	4.166	0.184	0.22	4.32
30 yr	1.045	1.582	0.262	0.66	3.65	0.817	3.084	0.206	0.26	3.53

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|\mathrm{SR}_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

4.2 Robustness: two-factor $\alpha + MKT$ models

Tables 8–9 and Figure 15 present the two-factor MKT + α model for BO and VC. The results show the instability anticipated by the simulation study: once an alpha intercept is introduced, separation between the intercept and the MKT loading weakens materially.

For BO, the alpha term absorbs a substantial share of the variation that otherwise loads on the market factor: under EW, $\hat{\alpha}$ ranges from 0.003 to 0.006 per month (and -0.001 for 17.5 years) and the MKT coefficient oscillates between 0.35 and 1.01, with no stable convergence across Horizons. Under FW, $\hat{\alpha}$ is persistently around 0.004–0.008 per month while the MKT loading falls to values between 0.11 and 0.44. The asymptotic standard errors for FW are highly erratic, with several entries at or very near zero interspersed with values exceeding 25. Cross-validation fold dispersions for $\hat{\beta}_{\mathrm{MKT}}$ are large (0.16–0.44 under EW; 0.64–0.74 under FW), indicating that these point estimates are highly sensitive to the sample split. Accordingly, the few isolated asymptotic significance markers in Tables 8–9 are not emphasized in our interpretation.

For VC, the two-factor model exhibits a different but similarly unstable pattern. At short Horizons (0 under EW), the alpha term reaches its upper bound ($\hat{\alpha} = 0.020$/month) and the MKT coefficient drops to approximately 0.47, resembling the BO behavior. However, as the Horizon increases beyond 5 years, the MKT loading rises steadily to values exceeding 2.3, while the alpha estimate turns progressively negative, reaching -0.009/month at 30 years. We interpret this reversal as further evidence that the two-parameter decomposition is highly sensitive to Horizon choice and sample composition in VC. Under FW, the instability is somewhat attenuated ($\hat{\alpha}$ remains close to zero at short Horizons; $\hat{\beta}_{\mathrm{MKT}} \approx 1.74$), but the overall fragility persists, with asymptotic standard errors exceeding 10 for some Horizons.

The BO and VC two-factor results therefore reinforce the same conclusion: among the specifications considered here, single-factor MKT models are the most stable for private-equity SDF estimation in our sample.

Table 8: Empirical Estimates: BO — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	1.004	0.845	0.157	1.19	6.04	0.152	28.325	0.698	0.01	0.20
2.5 yr	0.417	1.832	0.404	0.23	1.38	0.277	19.534	0.741	0.01	0.31
5 yr	1.003	2.739	0.333	0.37	2.54	0.398	5.088	0.695	0.08	0.52
10 yr	0.485	1.527	0.351	0.32	1.78	0.437	3.312	0.643	0.13	0.75
12.5 yr	0.348	5.339	0.417	0.07	1.39	0.348	5.102	0.641	0.07	0.69
15 yr	0.354	2.326	0.391	0.15	1.31	0.304	8.993	0.668	0.03	0.60
17.5 yr	0.995	2.466	0.394	0.40	1.85	0.301	14.401	0.699	0.02	0.58
20 yr	0.454	0.775	0.413	0.59	1.66	0.204***	0.046	0.726	4.41	0.49
25 yr	0.453	0.766	0.435	0.59	1.35	0.110	14.320	0.723	0.01	0.46
30 yr	0.453	0.766	0.435	0.59	1.35	0.110	14.320	0.723	0.01	0.46
Panel B: Alpha ($\hat{\alpha}$)										
0 yr	0.004	0.005	0.002	0.91	1.76	0.008	0.209	0.005	0.04	1.53
2.5 yr	0.006	0.018	0.003	0.35	1.78	0.007	0.141	0.005	0.05	1.28
5 yr	0.003	0.017	0.003	0.16	1.21	0.006	0.040	0.005	0.15	1.16
10 yr	0.004**	0.001	0.003	3.08	1.13	0.004	0.018	0.004	0.24	0.83
12.5 yr	0.004	0.093	0.003	0.05	0.95	0.004	0.036	0.004	0.12	0.78
15 yr	0.004	0.037	0.003	0.10	1.00	0.004	0.074	0.004	0.06	0.70
17.5 yr	-0.001	0.146	0.003	-0.01	0.28	0.004	0.124	0.005	0.03	0.58
20 yr	0.002	0.022	0.003	0.09	0.23	0.004***	0.000	0.005	9.79	0.59
25 yr	0.002	0.021	0.004	0.10	0.36	0.005	0.137	0.005	0.04	0.62
30 yr	0.002	0.021	0.004	0.10	0.36	0.005	0.137	0.005	0.04	0.62

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

Table 9: Empirical Estimates: VC — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	0.472	18.883	0.188	0.02	3.89	1.741	1.137	0.586	1.53	2.41
2.5 yr	1.426	1.813	0.365	0.79	3.76	1.563	8.719	0.343	0.18	4.62
5 yr	1.476	2.309	0.327	0.64	4.42	1.466	7.777	0.320	0.19	4.83
10 yr	1.801	26.027	0.182	0.07	10.02	1.888***	0.246	0.232	7.69	8.69
12.5 yr	1.977	4.756	0.194	0.42	10.21	2.010	10.488	0.153	0.19	13.64
15 yr	2.136	6.428	0.439	0.33	4.51	2.128	8.550	0.150	0.25	14.49
17.5 yr	2.233	5.744	0.228	0.39	9.67	2.173	21.351	0.188	0.10	11.71
20 yr	2.293	16.641	0.248	0.14	9.06	2.190	6.838	0.225	0.32	9.82
25 yr	2.315	4.671	0.250	0.50	9.04	2.191	5.138	0.235	0.43	9.34
30 yr	2.335	4.546	0.255	0.51	8.95	2.201	4.533	0.230	0.49	9.60
Panel B: Alpha ($\hat{\alpha}$)										
0 yr	0.020	0.115	0.009	0.17	1.63	-0.001	0.004	0.010	-0.35	0.44
2.5 yr	0.007	0.009	0.005	0.81	1.35	0.001	0.041	0.003	0.03	0.58
5 yr	0.006	0.012	0.004	0.53	1.15	0.002	0.037	0.004	0.05	0.34
10 yr	-0.001	0.123	0.002	-0.01	-1.29	-0.004***	0.001	0.001	-5.13	-4.28
12.5 yr	-0.004	0.007	0.001	-0.65	-5.73	-0.006	0.058	0.001	-0.11	-7.70
15 yr	-0.006	0.036	0.002	-0.18	-2.95	-0.008	0.050	0.001	-0.16	-7.45
17.5 yr	-0.007	0.033	0.001	-0.22	-10.55	-0.008	0.048	0.001	-0.17	-6.31
20 yr	-0.008	0.088	0.001	-0.09	-9.68	-0.008	0.045	0.002	-0.19	-5.45
25 yr	-0.008	0.030	0.001	-0.28	-9.08	-0.009	0.011	0.002	-0.75	-5.25
30 yr	-0.009	0.029	0.001	-0.29	-9.19	-0.009	0.011	0.002	-0.82	-5.48

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

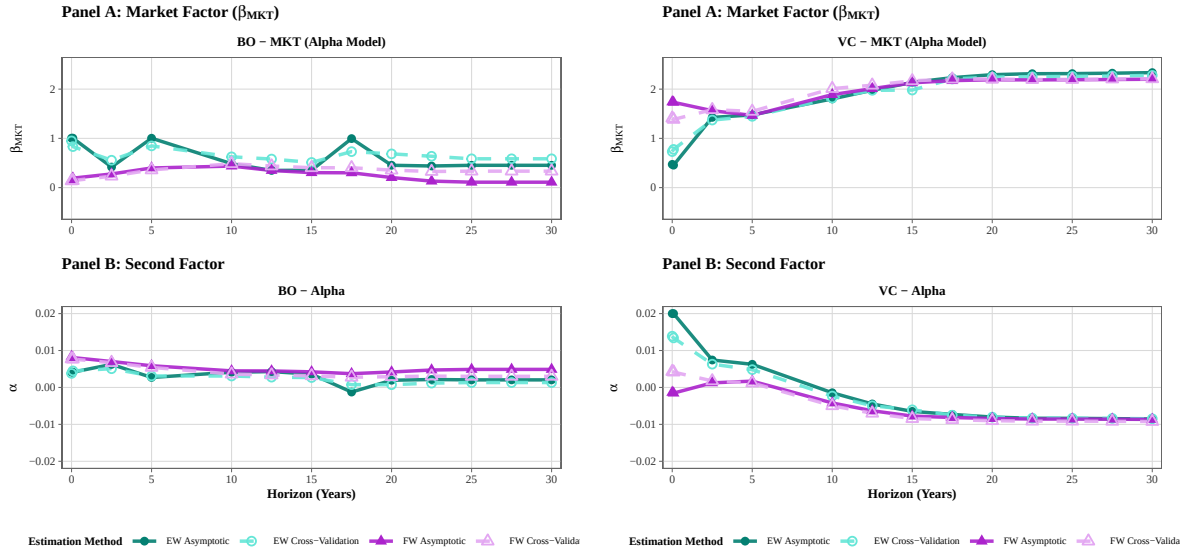

Figure 15: **Empirical ($MKT + \alpha$ model):** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

4.3 Two-factor q^5-factor models

We now pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG) in two-factor specifications estimated separately for BO and VC. Because the detailed estimation tables contain extensive Horizon-by-Horizon results, we do not reproduce them here and instead summarize the key patterns visible in Figures 16 and 17; the full tables are available in the accompanying online repository.

Buyout. Figure 16 shows a uniform pattern across the four non-market factors. At short Horizons (0–5 years), the second-factor loadings are positive and economically large: the ME loading reaches approximately 1.2–1.6 under EW (with stability ratios SR_{CV} of 3.0–3.8), the IA loading starts near 1.0 under EW ($\mathrm{SR}_{CV} \approx 1.4$), ROE near 0.6 ($\mathrm{SR}_{CV} \approx 0.9$), and EG near 0.5 ($\mathrm{SR}_{CV} \approx 1.3$). As the Horizon increases, however, all four second-factor loadings decay toward zero and frequently change sign beyond 10–15 years. The ME factor retains a positive loading above one even at long Horizons under FW ($\hat{\beta}_{\mathrm{ME}} \approx 1.2$, $\mathrm{SR}_{CV} \approx 2.1$), and under EW the loading drops to 0.9 after 17.5 years. In contrast, the EG, IA, and ROE loadings all fluctuate around zero at intermediate and long Horizons, with small absolute stability ratios relative to the single-factor benchmark. The MKT loading in these two-factor models remains similar to its single-factor counterpart (ranging from roughly 1.0–1.1 at short Horizons to 0.63–0.93 at long Horizons), showing that a second factor does not materially alter the market risk assessment. The asymptotic standard errors are erratic throughout, often exceeding 10 and occasionally reaching triple-digit values, which highlights the unreliability of asymptotic inference in this setting.

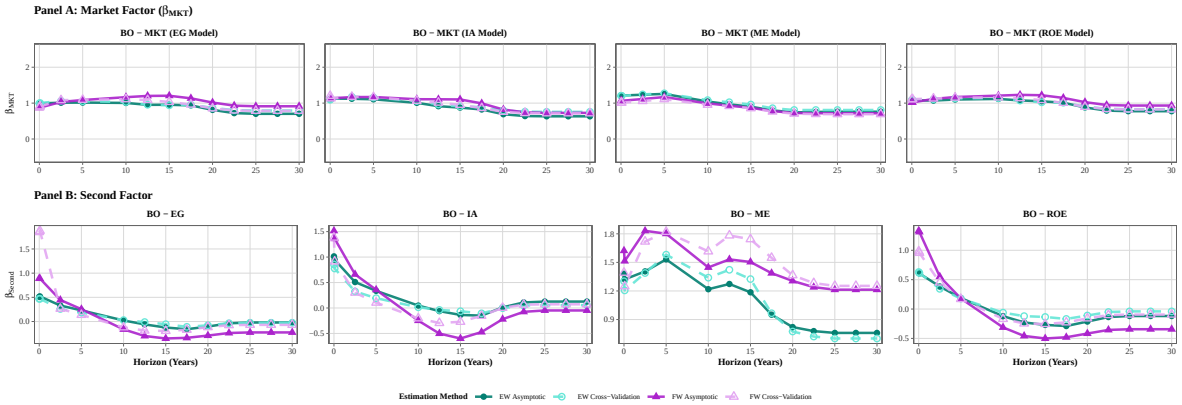

Figure 16: **Empirical results (q^5-factors):** Two-factor model for BO funds using VYP and maximum vintage year 2010.

Venture Capital. Figure 17 displays qualitatively analogous but more extreme patterns, consistent with the higher idiosyncratic risk inherent to early-stage ventures. At short Horizons, the second-factor loadings are strikingly large: the IA loading reaches 4.5 under EW at the 0-year Horizon ($\mathrm{SR}_{CV} \approx 1.4$) and 4.0 under FW, the ROE loading starts at 2.2 (EW) and 1.89 (FW), the EG loading at approximately 2.2 (EW) and 3.8 (FW), while the ME factor is the exception with a subdued loading near zero or slightly positive (0.38 under EW, −0.25 under FW). All four second-factor loadings undergo dramatic sign reversals as the Horizon extends. The IA loading, for instance, turns negative beyond 10 years and eventually reaches −1.5 to −1.9 at long Horizons, with correspondingly large absolute stability ratios ($|\mathrm{SR}_{CV}|$ exceeding 30). Similarly, EG and ROE produce large negative loadings at long Horizons ($\hat{\beta}_{\mathrm{EG}} \approx -1$, $\hat{\beta}_{\mathrm{ROE}} \approx -1.3$ under EW at 30 years). These sign reversals echo the behavior documented in the simulation study (Section 3.5), where two-factor models exhibited severe attenuation and sign changes in the second-factor loading across Horizons. The MKT loading in the VC two-factor models varies substantially across specifications: in the IA model it stabilizes around 1.3–1.5 at intermediate Horizons, while in the EG model it ranges from 0.4 to 2.0 (and between 1.0 and 1.9 for ROE) depending on the Horizon and weighting scheme.

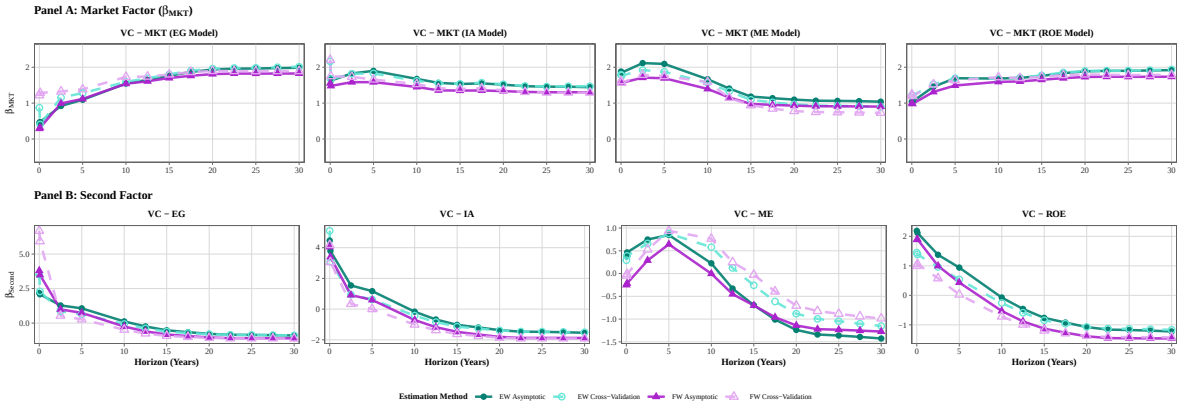

Figure 17: **Empirical results (q^5-factors):** Two-factor model for VC funds using VYP and maximum vintage year 2010.

Summary. The q^5-factor results for BO and VC confirm the fragility of multi-factor SDF estimation documented in the preceding MKT + α analysis. While a few second-factor loadings attain conventional asymptotic significance at isolated Horizons (particularly ME for BO and short-Horizon IA for VC), we do not treat those isolated t_A values as decisive because they are not stable across the Horizon grid and are not accompanied by comparably robust cross-validation evidence. None of the four non-market factors produces a stable, economically interpretable loading across the full Horizon spectrum. The pervasive sign reversals and the extreme sensitivity of point estimates to the compounding Horizon are more consistent with weak identification and Horizon sensitivity in the current sample than with a stably identified second factor. We therefore do not view the q^5-factor two-factor models as providing comparably stable evidence relative to the single-factor MKT benchmark for either asset class.

4.4 Interpretation

The empirical results for Buyout and Venture Capital funds are best interpreted as evidence about what current private-equity cash-flow panels can identify under semiparametric SDF estimation, rather than as definitive evidence about the true factor structure of private equity.

Among the specifications considered, the single-factor MKT model is the most stable. For both BO and VC, the single-factor market model delivers comparatively stable horizon-specific estimates across Horizons and weighting schemes, with magnitudes that remain economically interpretable. For Buyout, the estimated market beta declines from approximately 1.24 at short Horizons to 0.68 at long Horizons under equal weighting. For Venture Capital, the corresponding range is 1.88–2.00 down to 1.05–1.18, consistent with the higher systematic risk exposure typically associated with early-stage investments. Most importantly, the cross-validation stability diagnostics remain comparatively strong across the Horizon grid, even when asymptotic standard errors fluctuate erratically. In light of the simulation evidence and the limited

cross-section of vintage-year portfolios, we interpret the asymptotic results in this application as a useful large-sample benchmark, but not as decisive finite-sample inference.

Adding α or a second factor materially weakens identification. Once a second parameter is introduced, the empirical estimates become substantially less stable. In the MKT + α specification, the intercept and market loading are only weakly separated in the available sample, so that the estimated α often absorbs variation that is attributed to MKT in the single-factor model. This produces implausible or near-collapsed market betas and highly erratic uncertainty measures. In the q^5-based two-factor models, the second-factor loadings vary sharply across Horizons and often attenuate toward, or pass through, zero at longer Horizons. We view these sign changes and sensitivity patterns as evidence of weak finite-sample identification in the present data environment, not as definitive evidence that the underlying economic factors are absent from private-equity payoffs.

The BO–VC split strengthens the identification interpretation. Analyzing Buyout and Venture Capital separately is informative because the two segments differ markedly in business models, leverage, cash-flow timing, and payoff profiles. Despite these economic differences, both sub-samples exhibit the same broad empirical pattern: relatively stable single-factor market estimates, destabilization once α is introduced, and fragile second-factor estimates. This common pattern across BO and VC suggests that the central difficulty is not merely the pooling of heterogeneous fund types. At the same time, it does not imply that BO and VC share the same true factor structure; rather, it indicates that the present sample does not support stable estimation of richer specifications for either segment.

The empirical patterns are qualitatively consistent with the simulation study. The empirical findings line up closely with the mechanisms highlighted in Section 3.5. First, the Horizon range at which the single-factor estimates appear most stable, approximately 15–20 years, is close to the 12.5–15-year region that minimizes MSE in the simulations. Second, the attenuation and occasional sign reversals in the empirical two-factor estimates resemble the weak-identification patterns observed in the simulated multi-factor experiments. Third, the gap between unstable asymptotic standard errors and more informative stability diagnostics is consistent with the simulation result that asymptotic approximations can substantially mischaracterize uncertainty in small samples. Taken together, the empirical results suggest that the finite-sample mechanisms isolated in the simulations are economically relevant in real-world private-equity cash-flow data.

Robustness to loss specification. Private equity cash flows can be heavy-tailed, raising the concern that the baseline quadratic loss function ($L(x) = x^2$) might disproportionately weight outlier pricing errors. As a robustness test, we re-estimated our main specifications using the absolute deviation (L_1) loss function ($L(x) = |x|$). This alternative specification yields empirical results that closely mirror our baseline L_2 estimates in both magnitude and horizon-stability. This consistency suggests that the finite-sample instability documented above is driven by the structural identification challenges of pooling cash flows rather than extreme outlier sensitivity.

Implications for empirical practice. For applied work, the main lesson is methodological. In the present sample, semiparametric SDF estimation for private-equity cash flows appears most credible when it is restricted to seasoned vintage-year portfolios, emphasizes parsimonious market-based specifications, and evaluates robustness over a Horizon grid. Richer specifications with α terms or additional factors may still be economically meaningful, but the current data do not identify them with sufficient stability to support strong empirical conclusions. Accordingly, empirical claims in this setting should place greater weight on dependence-robust stability diagnostics, such as hv-block cross-validation, and less weight on formal asymptotic significance statements. A broader and more independent cross-section (through longer panels of seasoned funds, more observations within each vintage year, or both) would be needed before multi-factor semiparametric SDF estimates can be interpreted with comparable confidence.

5 Conclusion

Our generalized Least-Mean-Distance framework provides a flexible way to study identification and estimation of Stochastic Discount Factor (SDF) models using non-traded cash flows. The central theoretical result is that horizon choice is part of the estimand: inception-only discounting preserves the structural pooled pricing restriction, whereas horizon averaging can improve finite-sample stability while shifting the population target. Our Monte Carlo simulations show that this tradeoff is economically relevant in the reported designs. Short-Horizon criteria can be highly unstable, particularly when abnormal performance (α) is estimated jointly with market exposure. In contrast, averaging pricing errors across multiple dates can materially improve finite-sample recoverability through both lower dispersion and lower higher-order bias (even though the associated estimator generally targets a horizon-specific pseudo-true parameter rather than the structural inception-based benchmark).

In the data-sparse private-equity domain, where estimation relies on a limited cross-section of vintage-year portfolios, our empirical application shows that large-sample asymptotic approximations should be interpreted with caution. The asymptotic theory remains valuable as an organizing benchmark and as motivation for dependence-robust variance estimation, but in the empirical sample its implications are often dominated by finite-sample distortions. For that reason, we place greater weight on simulation evidence and on stability diagnostics tailored to overlapping cash-flow dependence (e.g., via hv-block cross-validation) than on formal asymptotic significance statements alone.

The empirical takeaway is therefore diagnostic rather than definitive. Current private-equity cash-flow panels appear informative about coarse market exposure in seasoned BO and VC portfolios, but substantially less informative about abnormal performance and richer factor decompositions. In our sample, single-factor MKT specifications are markedly more stable than models that add an α term or additional factors. We do not interpret this as evidence that private equity is truly priced by a single factor in population. Rather, we interpret it as evidence that, under currently available commercial cash-flow data and semiparametric pooled-cash-flow estimation, multi-parameter models are weakly identified and highly sensitive to finite-sample noise. Accordingly, parsimonious specifications should be preferred for empirical implementation unless materially broader cross-sectional information becomes available.

These results also bear on the literature on private-equity outperformance. Our findings do not imply that private-equity alpha is zero, nor do they imply that prior estimates of outperformance are necessarily incorrect. Rather, they indicate that semiparametric alpha estimates constructed from sparse pooled cash-flow panels can be weakly identified and highly sensitive to Horizon choice, sample construction, and finite-sample noise. This qualification is directly relevant for interpreting prior evidence on private-equity outperformance and risk. Differences across studies may reflect not only economic heterogeneity and dataset coverage, but also the limited ability of currently available cash-flow panels to separately identify abnormal performance from systematic risk exposure with high precision.

Expanding our SDF framework to robustly accommodate multiple risk factors requires imposing additional economic or statistical structure. Future research could further explore parametric assumptions for the cash flow generating process (Ang et al., 2018) or extract intra-period information from reported Net Asset Values (Brown et al., 2023). Methodologically, addressing the high estimator variance demands statistical techniques suited for sparse, high-dimensional settings. Approaches such as LASSO, ridge regression, or model averaging to regularize non-market coefficients, as well as ensemble learning methods that combine multiple weak learners, present promising paths forward. Especially resampling-based methods are likely to perform well in conjunction with Korteweg and Nagel (2016)-type loss functions, which aggregate pricing errors over multiple vintage years.

Another vital avenue for future work involves explicitly modeling funding rounds, financial engineering, complex fee structures and carried interest within the data generating process. This would enable the estimation of company-level and gross-versus-net-of-fee SDFs, providing sharper evaluation models for both General Partners (GPs) and Limited Partners (LPs). Such structural refinements would naturally improve parameter accuracy and allow for a more rigorous analytical characterization of finite-sample bias. Additionally, carefully integrating intermediate NAVs as proxies for unobservable true valuations could help alleviate the Compounding-Horizon Wedge inherent to terminal value concepts, though this requires downweighting actual cash distributions in favor of subjective accounting metrics.

Finally, we highlight the important role of the compounding Horizon in empirical asset pricing for private markets. Reporting estimates on a future-value-based horizon grid yields an additional array of alternative parameter estimates that can be compared to the traditional NPV-based results of the Driessen et al. (2012) method. Our simulation study shows that utilizing this future-value framework, i.e., averaging pricing errors over a broad set of discounting dates $\mathcal{T}$, can materially improve finite-sample stability relative to traditional net present value approaches in the reported designs. In the baseline simulations, bias is minimized at intermediate Horizons, around 12.5–15 years, rather than at the inception-only benchmark. Developing a formal theoretical or empirical mechanism to endogenously determine the optimal configuration of $\mathcal{T}$ for a given fund sample and macroeconomic environment remains an important objective for future research.

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Online appendices

Appendix A analyzes the target shift and finite-sample stabilization under $\mathcal{T}$-averaging. Appendix B compares our simulation and empirical results to their Driessen et al. (2012) counterparts. Appendix C provides the regularity conditions, inference details, and mathematical proofs.

A Target shift and finite-sample stabilization under $\mathcal{T}$-averaging

This appendix clarifies why $\mathcal{T}$-averaging can improve finite-sample performance even though, for $H > 0$, it generally shifts the population target away from the structural parameter θ_0. The key mechanism is a tradeoff between: (i) a *population target shift* induced by post-investment timing terms through the Compounding-Horizon Wedge, and (ii) a *finite-sample stabilization effect* from averaging imperfectly correlated pricing errors across discounting dates.

Throughout this appendix, we specialize to the quadratic loss $L(x) = x^2$ and set $w_i = 1$, so that the population objective reduces to a nonlinear least-squares criterion. Because this appendix works directly with squared loss rather than its negative, the horizon-specific population criterion is written in minimization form; this is equivalent to the maximization convention used for the negative-loss objective in Section 2.3. To keep the notation transparent, all moment calculations in this appendix are stated conditional on a fixed fund structure for a generic fund

i (that is, conditional on J_i, the relevant investment dates, and the horizon set $\mathcal{T}_{i,H}$), while expectations continue to integrate over SDF and pricing-error randomness. The corresponding unconditional population statements follow by taking an outer expectation over the distribution of fund structures.

To keep the different objects distinct, we proceed in four steps. First, we define the horizon-specific population criterion and pseudo-true minimizer. Second, we identify the exact source of the averaged population moment shift at θ_0. Third, we show how that moment shift enters the gradient of the population criterion and thereby affects the pseudo-true parameter. Fourth, we present an exact MSE decomposition and then interpret the simulation evidence through a second-order finite-sample lens. In compact form, the mechanism is: post-investment dates create timing-related covariance wedges at the deal level; these generate a nonzero averaged population moment $b_{H,i}$ at θ_0, tilt the gradient of the criterion, shift the pseudo-true target, and then enter the exact MSE tradeoff.

A.1 Setup and horizon-specific population target

For a given maximum compounding horizon H, let

$$\mathcal{T}_{i,H} := \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\},$$

where $T_i^{\max}$ denotes the largest discounting date available for fund i. Define the horizon-averaged pricing error by

$$\bar{\epsilon}_i^{(H)}(\theta) := \frac{1}{\operatorname{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta). \tag{21}$$

The associated population criterion is

$$Q_H(\theta) := \mathbb{E}\left[\left(\bar{\epsilon}_i^{(H)}(\theta)\right)^2\right], \tag{22}$$

and the horizon-specific pseudo-true parameter is defined by

$$\theta_H^\dagger := \arg\min_\theta Q_H(\theta), \tag{23}$$

whenever the minimizer exists and is unique. For notational simplicity, the conditioning on the fixed fund structure is suppressed in $Q_H(\theta)$ and in subsequent expectations. Thus, $\theta_H^\dagger$ is the target of the *population nonlinear least-squares criterion*. It is not defined as the zero of the averaged population moment, and those two objects need not coincide in general.

When $H = 0$, the estimator uses only the benchmark NPV discounting date $\tau = t_i^0$. Because no fund has cash flows before inception and every deal satisfies $t_i^0 \leq t_{i,j}^{\text{Inv}}$, part (1) of Lemma 1 implies that, under correct specification,

$$\mathbb{E}\left[\bar{\epsilon}_i^{(0)}(\theta_0)\right] = 0.$$

If, in addition, the benchmark population criterion $Q_0(\theta)$ is uniquely minimized at θ_0, then

$$\theta_0^\dagger = \theta_0.$$

For $H > 0$, the inclusion of post-investment timing terms generally perturbs the population criterion, so that

$$\theta_H^\dagger \neq \theta_0 \qquad \text{in general.}$$

It is convenient to define the horizon-specific averaged population moment at the structural parameter, conditional on the fund structure of a generic fund i:

$$b_{H,i} := \mathbb{E}\left[\bar{\epsilon}_i^{(H)}(\theta_0) \,\middle|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\text{Inv}}\}_{j=1}^{J_i}\right]. \tag{24}$$

The next subsection shows that, under correct specification, $b_{H,i}$ is generated exactly by the post-investment timing-related covariance wedges that aggregate into the Compounding-Horizon Wedge. The corresponding unconditional object is $\mathbb{E}[b_{H,i}]$.

A.2 Exact source of the averaged population moment at θ_0

Starting from the average pricing error in Equation (11) and the deal-level decomposition in Equation (2), we can write

$$\bar{\epsilon}_i^{(H)}(\theta_0) = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta_0) = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j}. \tag{25}$$

Taking expectations yields

$$b_{H,i} = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\text{Inv}}\}_{j=1}^{J_i}\right]. \tag{26}$$

For each (τ, j) term, use the identity

$$\mathbb{E}[XY] = \text{Cov}(X, Y) + \mathbb{E}[X]\mathbb{E}[Y]$$

with

$$X = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \qquad Y = \delta_{i,j}.$$

This gives

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j}\right] = \text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right) + \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\right] \mathbb{E}[\delta_{i,j}]. \tag{27}$$

Under correct specification, Equation (1) implies

$$\mathbb{E}[\delta_{i,j}] = \mathbb{E}\left[\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}]\right] = 0.$$

Hence only the covariance term remains.

Now consider $\tau \le t_{i,j}^{\text{Inv}}$. Because $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is then $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$-measurable,

$$\begin{aligned}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\,\delta_{i,j}\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\,\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right]\right] \\ &= 0.\end{aligned} \tag{28}$$

Since $\mathbb{E}[\delta_{i,j}] = 0$, Equation (28) also implies

$$\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right) = 0 \qquad \text{for all } \tau \le t_{i,j}^{\text{Inv}}.$$

We therefore obtain the following exact characterization.

Lemma 2 (Exact source of the averaged population moment at θ_0)**.** *Under correct specification,*

$$b_{H,i} = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbf{1}\{\tau > t_{i,j}^{\text{Inv}}\}\, \text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}, \delta_{i,j} \,\middle|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\text{Inv}}\}_{j=1}^{J_i}\right). \tag{29}$$

In particular, benchmark NPV terms contribute zero to $b_{H,i}$, and the averaged population moment at θ_0 is shifted only by post-investment timing terms.

Proof. The result follows from Equation (26), the decomposition in Equation (27), the fact that $\mathbb{E}[\delta_{i,j}] = 0$ under correct specification, and the iterated-expectations argument in Equation (28) for all pairs (τ, j) satisfying $\tau \le t_{i,j}^{\text{Inv}}$. □

Lemma 2 identifies the exact source of the averaged population moment shift at θ_0. It does not, by itself, characterize the pseudo-true target shift $\theta_H^\dagger - \theta_0$, because that parameter shift depends on how the moment shift enters both the gradient and the curvature of the population criterion in Equation (22).

Misspecification. In practice, all parsimonious SDF models are misspecified for observed PE cash flows, so the case $\mathbb{E}[\delta_{i,j}] \neq 0$ is the empirically relevant one. Under misspecification, the second term in Equation (27) no longer vanishes, and the conditional averaged population moment at θ_0 contains both (i) the timing-related covariance wedge and (ii) an additional mean component

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\right]\mathbb{E}[\delta_{i,j}].$$

This mean component reflects systematic deal-level mispricing that the SDF does not span, compounded by the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ which reweights the mispricing at each discounting date. Crucially, horizon averaging need not eliminate this additional component without further sign or decay restrictions: unlike the timing-related covariance wedge, it does not arise from the compounding structure and therefore does not share the same averaging properties. Accordingly, the clean correct-specification decomposition of $\mathcal{T}$-averaging into a target shift induced by

the Compounding-Horizon Wedge and a variance-reduction benefit no longer applies without qualification. Under misspecification, the population target shift reflects both the timing channel characterized by Lemma 2 and the misspecification channel, and the two channels may reinforce or offset each other depending on the sign and magnitude of $\mathbb{E}[\delta_{i,j}]$.

A.3 From averaged moment shift to pseudo-true target shift

Lemma 2 identifies the exact source of the conditional averaged population moment $b_{H,i}$ at θ_0. To connect that result to the parameter target, define

$$g_{i,H}(\theta) := \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta),$$

and assume that differentiation can be interchanged with expectation, so that the gradient of the population criterion can be expressed as an expectation of the sample-level score contributions. Then

$$\nabla_\theta Q_H(\theta) = 2\,\mathbb{E}\Big[\bar{\epsilon}_i^{(H)}(\theta)\, g_{i,H}(\theta)\Big]. \tag{30}$$

Evaluating at θ_0 and decomposing the expectation around the conditional mean $b_{H,i}$ yields

$$\nabla_\theta Q_H(\theta_0) = 2\,\mathbb{E}[b_{H,i}\, g_{i,H}(\theta_0)] + 2\,\mathbb{E}\Big[\big(\bar{\epsilon}_i^{(H)}(\theta_0) - b_{H,i}\big) g_{i,H}(\theta_0)\Big]. \tag{31}$$

The second term equals $2\,\mathrm{Cov}\big(\bar{\epsilon}_i^{(H)}(\theta_0),\, g_{i,H}(\theta_0)\big)$ and captures the residual co-movement between the averaged pricing error and its parameter sensitivity beyond the conditional mean shift.

Equation (31) makes two points precise. First, the conditional moment shift in $b_{H,i}$ induced by the Compounding-Horizon Wedge is one component of the criterion tilt at θ_0. Second, the pseudo-true target depends not only on that conditional mean shift, but also on the residual co-movement between the averaged pricing error and its parameter sensitivity.

Assume now that Q_H is twice continuously differentiable on an open convex neighborhood containing the line segment between θ_0 and $\theta_H^\dagger$, and that its Hessian is nonsingular throughout that set. Since $\theta_H^\dagger$ satisfies

$$\nabla_\theta Q_H(\theta_H^\dagger) = 0, \tag{32}$$

a first-order Taylor expansion around θ_0 gives

$$0 = \nabla_\theta Q_H(\theta_0) + \nabla_\theta^2 Q_H(\bar{\theta}_H)\,(\theta_H^\dagger - \theta_0), \tag{33}$$

for some intermediate point $\bar{\theta}_H$ between θ_0 and $\theta_H^\dagger$. Hence

$$\theta_H^\dagger - \theta_0 = -\big[\nabla_\theta^2 Q_H(\bar{\theta}_H)\big]^{-1} \nabla_\theta Q_H(\theta_0). \tag{34}$$

Combining Equations (31) and (34), the pseudo-true target shift is driven by the full gradient tilt of the population criterion at θ_0. Lemma 2 therefore identifies an exact channel through which the Compounding-Horizon Wedge moves the population target, but $b_{H,i}$ alone does not fully characterize $\theta_H^\dagger - \theta_0$ unless additional restrictions are imposed.

Remark 3 (A generic magnitude bound)**.** *For each post-investment term $\tau > t_{i,j}^{\mathrm{Inv}}$, Cauchy–Schwarz yields*

$$\left|\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right)\right| \leq \sigma\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right)\sigma(\delta_{i,j}).$$

Moreover, the averaging operator contributes the factor $1/\mathrm{card}(\mathcal{T}_{i,H})$ in Equation (29). Thus, large target shifts require either highly volatile SDF ratios, highly dispersed deal-level pricing errors, or strong alignment between these objects.

A.4 Exact MSE decomposition

Let $\hat{\theta}_H$ denote the sample extremum estimator associated with the sample analogue of Q_H. For any horizon H, the identity

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0) \tag{35}$$

holds exactly. Here, θ_0 is the structural parameter, $\theta_H^\dagger$ is the pseudo-true minimizer of the population criterion Q_H (Equation (23)), and $\hat{\theta}_H$ is its sample analogue. Taking squared norms and expectations over the sampling distribution gives the following decomposition.

Proposition 5 (Exact MSE decomposition around the structural parameter)**.** *For any horizon H,*

$$\begin{aligned}\mathbb{E}\left[\|\hat{\theta}_H - \theta_0\|^2\right] &= \|\theta_H^\dagger - \theta_0\|^2 + 2(\theta_H^\dagger - \theta_0)^\top \mathbb{E}\left[\hat{\theta}_H - \theta_H^\dagger\right] \\ &\quad + \left\|\mathbb{E}\left[\hat{\theta}_H - \theta_H^\dagger\right]\right\|^2 + \mathrm{tr}\left(\mathrm{Var}(\hat{\theta}_H)\right).\end{aligned} \tag{36}$$

Proof. Write

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0),$$

and expand the squared norm. Next, since $\theta_H^\dagger$ is nonrandom,

$$\mathrm{Var}(\hat{\theta}_H - \theta_H^\dagger) = \mathrm{Var}(\hat{\theta}_H),$$

so

$$\mathbb{E}\left[\|\hat{\theta}_H - \theta_H^\dagger\|^2\right] = \left\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\right\|^2 + \mathrm{tr}\left(\mathrm{Var}(\hat{\theta}_H)\right). \qquad \square$$

Proposition 5 separates four conceptually distinct objects:

1. the *population target shift* $\|\theta_H^\dagger - \theta_0\|^2$,
2. the *interaction term* $2(\theta_H^\dagger - \theta_0)^\top \mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]$, which links the target shift to the estimator's finite-sample bias around the pseudo-true parameter and can be either positive or negative,
3. the *squared finite-sample bias around the pseudo-true parameter* $\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\|^2$,
4. the *sampling variance* $\mathrm{tr}(\mathrm{Var}(\hat{\theta}_H))$.

This decomposition is exact. What remains model-dependent is how these terms move with H. From this point onward, the appendix no longer derives exact comparative statics in H; instead, it provides a second-order heuristic for why finite-sample stabilization can dominate over some range of horizons.

Heuristic link to second-order finite-sample bias. For nonlinear extremum estimators, standard higher-order expansions imply that

$$\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger] = O(n^{-1}),$$

with a constant that depends on higher-order moments of the score and on local curvature; see, e.g., Nagar (1959); Rilstone et al. (1996); Newey and Smith (2004); Bao and Ullah (2007). In the present setting, those constants are naturally affected by the volatility and dependence structure of the objective contributions. This suggests the following heuristic channel: if $\mathcal{T}$-averaging reduces the volatility of $\bar{\epsilon}_i^{(H)}(\theta)$ without introducing a large target shift, it can improve finite-sample performance even when $H > 0$.

A convenient representation is

$$\mathrm{Var}\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = a_{i,H}^\top \Sigma_{i,H}(\theta) a_{i,H}, \tag{37}$$

where $a_{i,H}$ is the $\mathrm{card}(\mathcal{T}_{i,H}) \times 1$ averaging vector with entries $1/\mathrm{card}(\mathcal{T}_{i,H})$, and $\Sigma_{i,H}(\theta)$ is the covariance matrix of the vector

$$\big(\epsilon_{\tau,i}(\theta)\big)_{\tau \in \mathcal{T}_{i,H}}.$$

Remark 4 (A simple variance-reduction benchmark)**.** *To build intuition for the stabilization mechanism, consider a stylized case. Suppose that all $m = \mathrm{card}(\mathcal{T}_{i,H})$ date-specific pricing errors have a common variance $\mathrm{Var}(\epsilon_{\tau,i}(\theta)) = \sigma^2(\theta)$ and that every pair of distinct dates shares the same correlation $\rho(\theta)$, with $\rho(\theta) \geq -1/(m-1)$ so that the covariance matrix is positive semidefinite. The variance of the averaged error then takes the form*

$$\mathrm{Var}\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = \sigma^2(\theta)\, \frac{1 + \rho(\theta)(m-1)}{m}.$$

Two polar cases anchor the interpretation. If the errors are independent ($\rho = 0$), this reduces to σ^2/m: the classical variance reduction from averaging m i.i.d. terms. If the errors are perfectly correlated ($\rho = 1$), the variance equals σ^2: averaging provides no benefit because each date carries exactly the same information. For any intermediate correlation $0 < \rho < 1$, the variance lies strictly between σ^2/m and σ^2, interpolating smoothly between the two extremes. Mild negative correlation ($\rho < 0$) pushes the variance below σ^2/m, as partial offsetting further stabilizes the average.

In summary, the averaged objective contribution has lower variance than a representative single-date moment whenever the date-specific errors are less than perfectly correlated. This is the basic channel through which horizon averaging can stabilize the sample criterion.

A.5 Interpretation and reconciliation with the simulation study

The simulation evidence in Section 3.5 can now be interpreted through Proposition 5.

At short horizons, the estimator relies on a small number of volatile pricing moments. This can make the nonlinear criterion unstable and can generate substantial finite-sample bias around the benchmark target. As H increases, averaging combines additional pricing moments that are correlated, but not perfectly so, and this can smooth the sample criterion and improve finite-sample behavior.

Against this benefit stands the population target shift characterized above. Once H is large enough to include post-investment timing terms, the criterion is generally no longer minimized at θ_0. The resulting target shift induced by the Compounding-Horizon Wedge can be economically meaningful, but under the baseline calibration it is modest relative to the large finite-sample distortions observed at very short horizons.

This yields the central interpretation of the simulation results: for small to intermediate horizons, the stabilization benefit of $\mathcal{T}$-averaging can dominate the incremental target shift. In the reported designs, this dominance reflects not only lower variance of the averaged criterion contributions, but also smaller higher-order finite-sample bias of the nonlinear extremum estimator around its horizon-specific pseudo-true parameter, so that overall error relative to the structural benchmark can initially fall before target shift eventually dominates. This entails, for sufficiently long horizons, the marginal gain from additional averaging may weaken while the scope for target shift grows. Accordingly, the MSE-minimizing horizon need not be $H = 0$ and need not be arbitrarily large. Instead, it is generally *interior* and *DGP-dependent*.

We do not claim that each MSE component from Proposition 5 is globally monotone in H without additional restrictions. In particular, the aggregate Compounding-Horizon Wedge may contain offsetting covariance terms, and the effect of averaging on estimator variance depends on the covariance structure of the pricing moments and on the local geometry of the nonlinear criterion. For that reason, the appendix establishes the exact decomposition and the exact source of the averaged population moment shift, while the existence and location of the empirically relevant interior optimum are documented in the simulation study rather than asserted as a universal theorem.

The role of $\mathcal{T}$-averaging is also related to two familiar ideas. First, from a GMM perspective, averaging combines multiple imperfectly correlated pricing moments, which can improve finite-sample stability when those moments contain useful information. Second, from the perspective of misspecified moments, post-investment timing terms can be viewed as informative but potentially biased moment components: they perturb the population target, yet may still lower overall MSE when the induced target shift is small relative to the finite-sample stabilization they provide.

The main message is not that the Compounding-Horizon Wedge "washes out" as H grows. Rather, $\mathcal{T}$-averaging trades off a population target shift against a potentially large reduction in finite-sample estimation error. Under the baseline DGP, the latter effect dominates over an economically relevant range of horizons, which is why the simulation study favors intermediate horizons rather than the NPV-only benchmark.

B Comparison to Driessen et al. (2012)

This section aligns our simulation and empirical designs with the original methodology of Driessen et al. (2012) to ensure direct comparability. Accordingly, we focus our analysis on estimates derived from NPV-based discounting (Horizon $H = 0$).

B.1 Comparison to simulation results

(Driessen et al., 2012, Table 1) report Monte Carlo results for $n = 50$ funds per vintage-year portfolio with monthly idiosyncratic standard deviation $\sigma_{\text{monthly}} = 0.3$, corresponding to roughly 104% annualized volatility under $\sqrt{12}$ scaling. In our base case (S1), we use only $n = 20$ funds per vintage, which is more realistic for older vintages and smaller private capital asset classes beyond Buyout and Venture Capital. Scenarios S2–S10 in Table 10 systematically vary the simulation design to identify which features drive the remaining discrepancy between our results and the Driessen et al. (2012) benchmark.

Several structural differences between the two simulation designs should be noted. First, Driessen et al. (2012) use a shifted lognormal distribution for the idiosyncratic error term (cf. their online appendix, Equation 9), which ensures that the error addend has a finite lower bound (approximately -1) and avoids the non-linear distortions that arise when normally distributed errors occasionally produce extreme negative total returns. Second, they employ a "full Monte Carlo" approach and simulate the total market return R_{MKT} from a shifted lognormal distribution calibrated to S&P 500 moments (1980–2003), rather than conditioning on the historical path. Third, their deal investment timing follows a deterministic schedule (three deals per year, evenly spaced across the five-year investment period), rather than random uniform draws.

Interpretation of results. Under our original design with normal errors and $\sigma = 30\%$ (S3), the market beta is severely attenuated ($\hat{\beta} = 0.62$) and the spurious alpha (0.68%/month) is an order of magnitude larger than the benchmark (0.05%). Switching to the shifted lognormal error distribution (S4) modestly improves the beta estimate to 0.71 and reduces the alpha bias to 0.57%. Adding simulated market returns (S5) yields no further improvement ($\hat{\beta} = 0.66$), and deterministic investment timing (S7) produces a similarly attenuated beta of 0.70 without substantially reducing alpha variance.

The most effective modification is shortening the maximum deal holding period from 10 years to 5 years (S8), which significantly improves the beta estimate ($\hat{\beta} = 0.91$) and reduces the alpha bias to 0.30% without requiring lower idiosyncratic volatility. Combining this holding-period restriction with deterministic investment timing (S9) yields a similar beta estimate ($\hat{\beta} = 0.88$) and alpha bias (0.32%), with the deterministic timing contributing only a modest additional effect. By limiting the compounding duration, both scenarios restrict the geometric growth of idiosyncratic errors, reducing the frequency of degenerate alpha estimates and partially restoring the accuracy of the market beta. Tightening the alpha upper bound $\bar{\alpha}$ also dramatically affects results: at $\bar{\alpha} = 1\%$ (S6), 32% of draws hit the bound, and beta is close to the benchmark ($\hat{\beta} = 0.87$), but at the cost of artificially constraining the alpha search space. Interestingly, Driessen et al. (2012) report in their footnote 8 that they use a much higher upper limit for α: "In our application, putting an upper bound of 100% per year suffices to avoid this numerical issue." Combining all modifications, i.e., holding-period restrictions, deterministic timing, and

a tighter alpha bound of $\bar{\alpha} = 1\%$ (S10), produces our closest approximation to the benchmark. Under this configuration, the beta estimate improves to 0.93 and the alpha bias falls to 0.27%, though 10% of draws still hit the alpha bound. However, even under this most favorable scenario (S10), our estimator still exhibits substantially more beta attenuation ($\hat{\beta} = 0.93$ vs. 0.98) and alpha bias (0.27% vs. 0.05%) than the Driessen et al. (2012) benchmark. This sensitivity reveals a fundamental tension in the two-factor model: the alpha term absorbs residual nonlinearity that the linear SDF cannot capture, and bounding it arbitrarily distorts the beta estimate.

Overall, the large remaining discrepancy between our simulation results and those of Driessen et al. (2012), even after adopting their shifted lognormal errors and simulated market returns, implies that we were unable to reproduce the favorable small-sample properties reported in Driessen et al. (2012). Across all scenarios with $\sigma = 30\%$, the beta estimates range from 0.62 to 0.93, well below the Driessen et al. (2012) benchmark of 0.98, confirming the estimator's sensitivity to the interplay between idiosyncratic noise, compounding Horizon, and alpha bounds.

Table 10: Comparison to simulation (Driessen et al., 2012, Table 1)

	Market Beta (β)					Alpha (α, monthly %)					
Scenario	True	Mean	SD	25%	75%	True	Mean	SD	25%	75%	% Bound
(S1) $n = 20$, $\sigma = 20\%$, Normal	1.00	0.84	0.40	0.61	1.05	0.00	0.22	0.28	0.04	0.32	0%
(S2) $n = 50$, $\sigma = 20\%$, Normal	1.00	0.89	0.29	0.74	1.03	0.00	0.13	0.16	0.03	0.19	0%
(S3) $n = 50$, $\sigma = 30\%$, Normal	1.00	0.62	0.72	0.12	1.03	0.00	0.68	0.70	0.13	1.08	0%
(S4) $n = 50$, $\sigma = 30\%$, Shifted LN	1.00	0.71	0.72	0.23	1.12	0.00	0.57	0.67	0.07	0.91	0%
(S5) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT	1.00	0.66	0.79	0.11	1.06	0.00	0.71	0.79	0.03	1.34	0%
(S6) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, $\bar{\alpha} = 1\%$	1.00	0.87	0.69	0.41	1.24	0.00	0.48	0.50	0.06	1.00	32%
(S7) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Det. Timing	1.00	0.70	0.79	0.19	1.12	0.00	0.65	0.80	-0.01	1.20	0%
(S8) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5	1.00	0.91	0.54	0.64	1.19	0.00	0.30	0.52	-0.03	0.47	0%
(S9) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing	1.00	0.88	0.54	0.61	1.16	0.00	0.32	0.51	-0.01	0.53	0%
(S10) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing, $\bar{\alpha} = 1\%$	1.00	0.93	0.50	0.65	1.17	0.00	0.27	0.39	-0.01	0.52	10%
Driessen et al. 2012, Table 1											
Benchmark Estimates ($n = 50$, $\sigma_{\text{monthly}} = 30\%$)	1.00	0.98	0.32	0.80	1.15	0.00	0.05	0.2	-0.06	0.15	n/a

Notes: This table reports the small-sample properties of the generalized estimator for Horizon 0 (i.e., treating funds as held to maturity), mimicking the structure of Table 1 in Driessen et al. (2012). We simulate 1,000 Monte Carlo draws. All scenarios use 20 vintage years (1986–2005) with 15 deals per fund; unless stated otherwise, $n = 20$ funds per vintage, $\sigma = 0.2$ monthly idiosyncratic volatility, and a normal error distribution. The simulation dimensions vary as follows: n = number of funds per vintage-year portfolio; σ = monthly idiosyncratic standard deviation; *Normal* = Gaussian error term (default); *Shifted LN* = shifted lognormal error term (cf. Driessen et al. (2012)); *Sim. MKT* = total market return drawn from a shifted lognormal (S&P 500 1980–2003 moments) with a constant risk-free rate of 4% annualized, instead of conditioning on the historical path; *Det. Timing* = deterministic (evenly spaced) deal investment dates instead of random; *Max. Hold 5* = maximum holding period of a deal limited to 5 years (default 10 years); $\bar{\alpha}$ = upper bound on the alpha parameter in the optimization (default $\bar{\alpha} = 2\%$/month). The *% Bound* column reports the fraction of draws where the alpha estimate reached the upper optimization bound $\bar{\alpha}$. The final row presents the Benchmark Estimates obtained directly from Table 1 in Driessen et al. (2012) for comparative purposes. Alpha estimates are monthly and presented in percentages.

B.2 Comparison to empirical results

In this section, we evaluate the empirical stability of single- and multi-factor SDF specifications by estimating these models on our dataset (for the US-Dollar North America subsample with maximum vintage 2010) and comparing the results directly with the benchmark estimates published by Driessen et al. (2012). A key distinction is the underlying private market dataset:

we use the Preqin cash flow database, which is also employed by Korteweg and Nagel (2016) and Ang et al. (2018), whereas Driessen et al. (2012) rely on the Thomson Venture Economics (TVE) fund-level dataset, which suffers from well-documented data quality issues such as forward-filling of stale KPIs (Harris et al., 2014). The cash flow summary of (Driessen et al., 2012, Table 3) indicates fund-value weighted vintage-year portfolios. (Driessen et al., 2012, Table 5) report results for Buyout (BO) and Venture Capital (VC) funds using both a two-factor model (S&P 500 MKT + α) and the Fama and French (1993) four-factor model (MKT, SMB, HML, and α). In their two-factor specification, VC exhibits a market beta of 2.73 and a highly negative annualized alpha of -12.3%, while BO yields a market beta of 1.31 and an annualized alpha of -4.8%. These strongly negative alpha estimates can be at least partially attributed to the aforementioned data errors in TVE. Indeed, the BO alpha coefficient is statistically insignificant once bootstrap standard errors are taken into account, whereas the VC alpha remains significant. In their four-factor specification, the factor loadings are broadly plausible: BO exhibits $\hat{\beta}_{\text{MKT}} = 0.72$, $\hat{\beta}_{\text{SMB}} = 0.50$, and $\hat{\beta}_{\text{HML}} = 0.19$, while VC shows $\hat{\beta}_{\text{MKT}} = 2.20$, $\hat{\beta}_{\text{SMB}} = 1.83$, and $\hat{\beta}_{\text{HML}} = -0.48$. However, Driessen et al. (2012) note that the four-factor model does not significantly improve upon the two-factor specification and that the SMB and HML loadings are largely insignificant.

Table 11 and Table 12 present our empirical results for the single-factor (MKT), two-factor (α, MKT), three-factor (MKT, SMB, HML), and four-factor (α, MKT, SMB, HML) Fama and French (1993) models for BO, VC, and PE at Horizon 0, using the conservatively restricted sample with a maximum vintage-year cutoff of 2010.

We begin with the single-factor MKT model to establish a stable baseline. Under this specification, market betas are consistently positive and theoretically plausible: the cross-validation fold averages ($\hat{\beta}_{\text{MKT}}$) are 1.21 for BO, 1.78 for VC, and 1.21 for PE under equal weighting (EW). The CV standard deviations verify that these estimates are highly stable across data subsets (e.g., CV standard deviations of 0.12 for BO and 0.09 for PE under EW). This result provides stable sample evidence that private equity cash flows co-move with the broader market.

However, when explicitly allowing for an abnormal performance α intercept alongside the market factor, the estimation becomes substantially less stable. In the two-factor model (α, MKT), the alpha parameter becomes highly sensitive to random, fund-specific noise compounded over a fund's lifetime. Even when restricting the sample to mature vintages, private equity datasets contain relatively few funds per year. As a result, the effective cross-section is limited relative to the dimensionality of the nonlinear criterion. Consequently, the optimizer often assigns inflated, positive alpha premiums (often hitting or nearing the 2% monthly bound, e.g., $\hat{\alpha} = 0.020$ for VC under EW and 0.008 for BO under FW), while the market betas are driven toward zero (e.g., $\hat{\beta}_{\text{MKT}} = 0.14$ for BO and 0.17 for PE under FW). While asymptotic standard errors correspondingly explode or round to zero, the CV standard deviations reveal the actual parameter variation across data subsets. For instance, the cross-validation fold dispersion for the MKT estimate exceeds 0.6 for both BO and PE under FW, confirming that the point estimates are highly sensitive to the sample and do not support strong interpretation.

This instability pattern extends directly to the multi-factor models. When estimating the three-factor model (MKT, SMB, HML) *without* an alpha term, the estimates roughly mirror the MKT-only model (e.g., BO $\hat{\beta}_{\text{MKT}} = 1.33$ under EW), and the CV analysis shows reasonable out-of-sample stability, even though the asymptotic standard errors are vast. The SMB loading

is consistently positive for BO and PE, but predominantly negative or near zero for VC. Notably, this sign for VC generally *reverses* the pattern reported by Driessen et al. (2012), who find a strongly positive VC SMB loading ($\hat{\beta}_{\text{SMB}} = 1.83$). The discrepancy likely reflects differences in the underlying datasets (Preqin vs. TVE) and sample periods, underscoring the sensitivity of multi-factor estimates in private equity.

But once the alpha term is reintroduced in the four-factor model (α, MKT, SMB, HML), the same instability reappears. Because the alpha term absorbs the bulk of the remaining variance, the other factor loadings swing erratically: the asymptotic MKT coefficient for BO drops substantially (to 0.15 under FW), while SMB and HML loadings reach extreme values. The pervasive asymptotic standard errors of exactly (0.000) in these models often occur when estimates lie on or very near the imposed parameter bounds. In such cases, the usual interior Hessian-based asymptotic approximation is not a reliable measure of uncertainty, so we interpret the SHAC standard errors with caution and place greater weight on the cross-validation stability diagnostics. The CV analysis exposes this reality: rather than being estimated with zero standard errors, the parameters swing erratically across hold-out sets, with CV standard deviations for the factor loadings routinely exceeding 0.4, indicating that the multi-factor models are not stably estimated when alpha is included.

In conclusion, our empirical evaluation is consistent with the simulation-based warning from Section 3.5: with the limited cross-section of only 20–40 vintage-year portfolios currently available in commercial private equity databases, SDF models that attempt to separately estimate multiple pricing factors alongside an explicit alpha parameter appear too weakly identified in this sample to support stable inference. The inclusion of an unconstrained or loosely constrained abnormal performance term confounds the signal with cross-sectional noise and materially destabilizes the factor model estimates. Therefore, until substantially more granular and independent cross-sectional data becomes cleanly available, the single-factor market model remains the most stable specification among those we estimate for private-equity SDF performance evaluation.

Table 11: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Buyout and Venture Capital

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Buyout (BO)								
1-Factor (Asymp.)		1.246				1.228		
		(1.195)				(1.169)		
1-Factor (CV)		1.214				1.168		
		(0.118)				(0.142)		
2-Factor (Asymp.)	0.004	1.004			0.008	0.175		
	(0.004)	(0.845)			(0.194)	(26.737)		
2-Factor (CV)	0.005	0.834			0.008	0.141		
	(0.003)	(0.336)			(0.005)	(0.712)		
3-Factor (Asymp.)		1.376	1.308	0.409		1.287	1.873	0.267
		(1.958)	(6.093)	(1.368)		(2.459)	(6.977)	(2.929)
3-Factor (CV)		1.333	1.129	0.375		1.224	1.521	0.161
		(0.171)	(0.378)	(0.144)		(0.157)	(0.544)	(0.290)
4-Factor (Asymp.)	0.007	0.457	0.730	-0.271	0.008	0.150	0.545	-0.008
	(0.005)	(0.000)	(0.000)	(0.000)	(0.008)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.003	1.007	0.064	0.078	0.008	0.272	0.381	-0.256
	(0.002)	(0.007)	(0.162)	(0.199)	(0.008)	(0.855)	(0.481)	(0.760)
Venture Capital (VC)								
1-Factor (Asymp.)		1.886				1.591		
		(1.775)				(1.666)		
1-Factor (CV)		1.779				1.594		
		(0.397)				(0.357)		
2-Factor (Asymp.)	0.020	0.466			-0.001	1.745		
	(0.121)	(19.898)			(0.003)	(0.835)		
2-Factor (CV)	0.014	0.731			0.004	1.416		
	(0.009)	(0.187)			(0.010)	(0.590)		
3-Factor (Asymp.)		2.032	-0.368	1.432		1.968	-0.454	1.406
		(0.757)	(2.189)	(0.454)		(0.834)	(1.649)	(1.552)
3-Factor (CV)		1.928	-0.343	1.218		1.896	0.045	1.122
		(0.449)	(0.444)	(0.963)		(0.668)	(0.727)	(1.468)
4-Factor (Asymp.)	0.004	1.626	-0.642	1.672	-0.001	1.969	-0.530	1.177
	(0.001)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.011	1.059	-0.270	1.257	0.004	1.770	-0.058	1.340
	(0.008)	(0.344)	(0.582)	(1.207)	(0.010)	(0.576)	(0.714)	(1.964)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

Table 12: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Private Equity

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Private Equity (PE)								
1-Factor (Asymp.)		1.208				1.160		
		(0.770)				(1.175)		
1-Factor (CV)		1.206				1.104		
		(0.091)				(0.137)		
2-Factor (Asymp.)	0.003	1.003			0.008	0.210		
	(0.002)	(0.401)			(0.499)	(75.481)		
2-Factor (CV)	0.004	0.873			0.007	0.174		
	(0.002)	(0.256)			(0.004)	(0.627)		
3-Factor (Asymp.)		1.246	0.790	0.458		1.202	1.690	0.213
		(2.365)	(1.505)	(1.247)		(3.696)	(4.854)	(2.065)
3-Factor (CV)		1.194	0.494	0.534		1.156	1.398	0.142
		(0.205)	(0.310)	(0.193)		(0.135)	(0.445)	(0.234)
4-Factor (Asymp.)	0.003	1.003	0.003	0.005	0.004	0.626	0.498	0.389
	(0.001)	(0.000)	(0.000)	(0.000)	(0.007)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.004	0.877	0.008	0.019	0.007	0.430	0.095	-0.147
	(0.002)	(0.261)	(0.048)	(0.121)	(0.005)	(0.486)	(1.014)	(0.741)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

C Asymptotic conditions, inference details, and proofs for the horizon-specific results

This appendix collects the full regularity conditions, the auxiliary uniform law of large numbers, the implementation details for large-sample inference, and the proofs omitted from Section 2.3. Throughout, fix a horizon $H \geq 0$.

C.1 Asymptotic setup and assumptions

The large-sample benchmark is built around the horizon-averaged pricing error

$$\bar{\epsilon}_i^{(H)}(\theta) := w_i \cdot \frac{1}{\mathrm{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta)$$

with

$$\mathcal{T}_{i,H} = \left\{ t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max}) \right\},$$

and the corresponding loss contribution

$$Z_i(\theta) := L\left(\bar{\epsilon}_i^{(H)}(\theta)\right), \qquad Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} Z_i(\theta).$$

By the value-additivity principle (Hansen and Richard, 1987), the SDF restriction can be formulated for pooled cash flows at different aggregation levels. In the empirical implementation, we form vintage-year portfolios as cross-sectional units, because this reduces idiosyncratic fund-level noise while preserving economically meaningful variation across macro environments.

Assumption 2. *For each vintage year v, we pool fund cash flows to form n_v portfolios, so that*

$$n = \sum_{v=1}^{V} n_v.$$

The boundary cases are: (i) single-fund portfolios and (ii) one portfolio per vintage year. Both equal-weighted and value-weighted portfolio constructions are admissible.

Without loss of generality, we continue to refer to the resulting cross-sectional units as funds. Let $v_i := \mathrm{YEAR}(t_i^0)$ denote the vintage year of unit i. The pricing error $\bar{\epsilon}_i^{(H)}(\theta)$ is therefore indexed by both unit identity and vintage year, so the data are naturally represented as an increasing-domain random field rather than as a standard iid cross-section or a conventional long time series. Economically, dependence arises because adjacent vintage years are exposed to overlapping macroeconomic conditions and partially overlapping realized cash-flow windows.

Assumption 3 (Vintage-year increasing-domain asymptotics)**.** *As $n \to \infty$:*

1. *The number of distinct vintage years satisfies $V \to \infty$.*
2. *The number of units per vintage year is uniformly bounded: there exists $\bar{n} < \infty$ such that $n_v \leq \bar{n}$ for all v.*
3. *Fund lifetimes are uniformly bounded: there exists $\bar{T} < \infty$ such that $T_i^{\max} - t_i^0 \leq \bar{T}$ for all i.*
4. *The economic distance between units i and j is measured by*

$$d_{i,j} = |v_i - v_j| + \rho_0 \mathbf{1}_{\{i \neq j\}}, \qquad \rho_0 > 0.$$

Assumption 3 imposes increasing-domain asymptotics and rules out infill asymptotics in the sense of Jenish and Prucha (2012). The role of $\rho_0 > 0$ is to prevent the field from becoming asymptotically dense at a fixed location. This is economically natural here: asymptotic information is generated by observing additional vintage years, not by observing an ever denser sample from a fixed macroeconomic environment.

A useful implication of Assumption 3(2) is that the total number of observations n and the number of vintages V are of the same order. Indeed,

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n}V,$$

so $n \asymp V$. Consequently, $\sqrt{n}$- and $\sqrt{V}$-rates are asymptotically equivalent up to a constant factor. Put differently, statistical information grows with the span of vintage years rather than with within-vintage clustering.

Whenever convenient, re-index the observations by vintage year and within-vintage slot. That is, for vintage year $v = 1, \ldots, V$ and slot $\ell = 1, \ldots, n_v$, let $Z_{v,\ell}(\theta)$ denote the loss contribution of the ℓth unit in vintage year v. By Assumption 3(2), $n_v \leq \bar{n} < \infty$ uniformly in v. Hence the sample can be embedded in the increasing strip

$$\Lambda_V := \{(v, \ell) : v = 1, \ldots, V,\ \ell = 1, \ldots, \bar{n}\},$$

where empty slots are padded with zeros. Because $\bar{n}$ is fixed while $V \to \infty$, this embedding preserves the order of the sample size, and Assumption 3(2) implies

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n}V, \qquad \text{so that} \qquad n \asymp V.$$

This increasing-domain perspective differs from the infill framework of Driessen et al. (2012), who fix the number of portfolios and let the number of underlying funds per portfolio diverge. Their asymptotic device is appropriate for consistency conditional on a fixed macroeconomic environment. By contrast, the present framework is designed for unconditional identification and inference when new vintages generate new realizations of macroeconomic conditions.

Assumption 4 (Parameter space)**.** *The parameter space $\Theta \subset \mathbb{R}^p$ is compact.*

Assumption 5 (Primitive regularity conditions for uniform convergence)**.** *For each fixed H:*

1. *The array*

$$\left\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1, \ldots, n\right\}$$

   is near-epoch dependent (NED) on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$, uniformly in $\theta \in \Theta$ (that is, the NED approximation constants are bounded uniformly over $\theta \in \Theta$ and i).

2. *The loss function $L(\cdot)$ is continuous and Lipschitz on the relevant support.*

3. *There exists $p > 2$ such that*

$$\sup_{\theta\in\Theta} \mathbb{E}\left[\left|L\left(\bar{\epsilon}_i^{(H)}(\theta)\right)\right|^p\right] < \infty$$

   uniformly in i.

4. *The criterion is stochastically equicontinuous in θ.*

Assumption 5 is stated at a deliberately primitive level: dependence is imposed on the underlying horizon-specific pricing errors, and the loss inherits the required regularity through continuity and Lipschitz continuity. This is preferable to assuming a law of large numbers directly for the transformed criterion because it ties the statistical assumptions more closely to the economic structure of the data.

Assumption 6 (Population stabilization across vintages)**.** *For each fixed H, the sample-specific mean criterion converges uniformly to the population criterion defined in Equation* (15)*:*

$$\sup_{\theta\in\Theta}\left|-\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\big[Z_i(\theta)\big]-Q_H(\theta)\right|\to 0.$$

Assumption 6 makes explicit the population-stabilization step across expanding vintage spans that was only implicit in the compact main-text notation: as the sample adds vintages, the Cesàro average of the expected criterion contributions converges to a single horizon-specific population object.

Assumption 7 (Identification of the horizon-specific target)**.** *For each fixed H, the population criterion $Q_H(\theta)$ has a unique maximizer*

$$\theta_H^{\dagger} := \arg\max_{\theta\in\Theta} Q_H(\theta).$$

Assumption 8 (Local Hessian convergence)**.** *For each fixed H, there exists an open neighborhood $\mathcal{N}_H$ of $\theta_H^{\dagger}$ such that*

$$\sup_{\theta\in\mathcal{N}_H}\left\|\nabla_{\theta\theta}Q_{n,H}(\theta)-\nabla_{\theta\theta}Q_H(\theta)\right\|\xrightarrow{p}0.$$

Assumption 8 is the standard sample-Hessian convergence condition used in extremum-estimator central limit theorems (e.g., Newey and McFadden, 1994). In our increasing-domain framework, it follows from the same random-field LLN argument as Proposition 6, provided that the Hessian contributions inherit the required NED, moment, and stochastic-equicontinuity properties from the pricing errors $\bar{\epsilon}_i^{(H)}(\theta)$ and the loss function $L(\cdot)$.

Together, these conditions supply the formal underpinning for the compact benchmark stated in Section 2.3. They pin down the population object associated with each fixed horizon, justify the increasing-domain normalization indexed by vintage years, and provide the regularity needed for the SHAC-based large-sample covariance approximation.

C.2 Auxiliary uniform law of large numbers

The first technical step is a uniform law of large numbers for the horizon-specific sample criterion $Q_{n,H}(\theta)$.

Proposition 6 (Uniform law of large numbers)**.** *Under Assumptions 2, 3, 5, and 6,*

$$\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - Q_H(\theta)\right| \xrightarrow{p} 0.$$

A short route to Proposition 6 is to apply a uniform law of large numbers for NED random fields on α-mixing bases; see, for example, Jenish and Prucha (2009, Theorem 6). The bounded-lifetime and bounded-within-vintage assumptions ensure that cross-sectional overlap remains local in vintage-year distance, while the moment and equicontinuity conditions deliver uniform stochastic control over $\theta \in \Theta$.

Economically, Proposition 6 says that when the sample spans sufficiently many distinct vintages, the empirical criterion stabilizes around its population counterpart. This remains true whether the horizon is structural ($H = 0$) or regularized ($H > 0$); what changes across horizons is the population target itself.

Proof of Proposition 6. Let

$$\bar{Q}_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\big[Z_i(\theta)\big].$$

It suffices to show that

$$\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - \bar{Q}_{n,H}(\theta)\right| \xrightarrow{p} 0, \qquad \sup_{\theta\in\Theta}\left|\bar{Q}_{n,H}(\theta) - Q_H(\theta)\right| \to 0,$$

because then the triangle inequality yields the claim.

We first verify the stochastic part. By Assumption 5(1), the array $\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1, \ldots, n\}$ is uniformly NED on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$. Since $L(\cdot)$ is continuous and Lipschitz by Assumption 5(2), $Z_i(\theta) = L(\bar{\epsilon}_i^{(H)}(\theta))$ inherits the same NED structure, uniformly in $\theta \in \Theta$. Moreover, Assumption 5(3) gives a uniform L^p envelope for some $p > 2$, and Assumption 5(4) gives stochastic equicontinuity in θ. Re-indexing by (v, ℓ) therefore produces an array of random fields on the strip Λ_V satisfying the primitive conditions of Jenish and Prucha (2009, Theorem 6). Assumption 3(2)–(4) ensures that the field is observed over an increasing domain, not an infill domain: the width of the strip is fixed, dependence is indexed by the vintage-year distance, and bounded lifetimes prevent the effective dependence neighborhoods induced by overlapping cash-flow windows from expanding with n. Hence

$$\sup_{\theta\in\Theta}\left|\frac{1}{n}\sum_{i=1}^{n}\Big(Z_i(\theta) - \mathbb{E}[Z_i(\theta)]\Big)\right| \xrightarrow{p} 0.$$

Equivalently,

$$\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - \bar{Q}_{n,H}(\theta)\right| \xrightarrow{p} 0.$$

It remains to pass from the sample-specific mean criterion $\bar{Q}_{n,H}(\theta)$ to the population target $Q_H(\theta)$ defined in Equation (15). Assumption 6 gives exactly this deterministic uniform convergence. Therefore,

$$\sup_{\theta\in\Theta}\left|\bar{Q}_{n,H}(\theta) - Q_H(\theta)\right| \to 0.$$

Combining the last two displays proves

$$\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - Q_H(\theta)\right| \xrightarrow{p} 0.$$

□

C.3 Proof of consistency

Proposition 2 is the horizon-indexed analogue of the standard extremum-estimator consistency result of Newey and McFadden (1994, Theorem 2.1). Proposition 6 supplies the required uniform convergence, Assumption 7 provides a unique maximizer, and compactness of Θ together with continuity of the pricing-error map delivers the remaining regularity.

Proof of Proposition 2. We verify the hypotheses of the standard argmax theorem directly.

Step 1: continuity of the population criterion. For each unit i, the map $\theta \mapsto \bar{\epsilon}_i^{(H)}(\theta)$ is continuous on Θ because $\mathcal{T}_{i,H}$ is finite and the discount-factor ratios $\Psi_{\tau,t}(\theta)$ are continuous in θ. Since $L(\cdot)$ is continuous, $\theta \mapsto Z_i(\theta)$ is continuous as well. Assumption 5(3) supplies an integrable envelope uniformly in i and θ, so dominated convergence implies that

$$Q_H(\theta) = -\mathbb{E}\left[Z_i(\theta)\right]$$

is continuous on Θ. Because Θ is compact by Assumption 4, Q_H attains its maximum on Θ; by Assumption 7, this maximizer is unique and equals $\theta_H^\dagger$.

Step 2: uniform convergence. Proposition 6 yields

$$\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - Q_H(\theta)\right| \xrightarrow{p} 0.$$

Step 3: argmax identification. Because $\hat{\theta}_H$ maximizes $Q_{n,H}$ over Θ, for every $\theta \in \Theta$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta).$$

In particular, with $\theta = \theta_H^\dagger$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta_H^\dagger).$$

Subtracting and adding the population criterion gives

$$\begin{aligned}
0 &\leq Q_H(\theta_H^\dagger) - Q_H(\hat{\theta}_H) \\
&\leq \left|Q_H(\theta_H^\dagger) - Q_{n,H}(\theta_H^\dagger)\right| + \left|Q_{n,H}(\hat{\theta}_H) - Q_H(\hat{\theta}_H)\right| \\
&\leq 2\sup_{\theta\in\Theta}\left|Q_{n,H}(\theta) - Q_H(\theta)\right|,
\end{aligned}$$

which converges to zero in probability. Hence

$$Q_H(\hat{\theta}_H) \xrightarrow{p} Q_H(\theta_H^\dagger).$$

Now let $\{n_k\}$ be any subsequence. By compactness of Θ, there exists a further subsequence, still denoted $\{n_k\}$ for notational simplicity, such that $\hat{\theta}_H \to \bar{\theta}$ almost surely along that subsequence for some random $\bar{\theta} \in \Theta$. By continuity of Q_H,

$$Q_H(\hat{\theta}_H) \to Q_H(\bar{\theta}) \qquad \text{almost surely along the subsequence.}$$

Since the same subsequence satisfies $Q_H(\hat{\theta}_H) \to Q_H(\theta_H^\dagger)$ in probability, it follows that $Q_H(\bar{\theta}) = Q_H(\theta_H^\dagger)$ almost surely. By uniqueness of the maximizer, $\bar{\theta} = \theta_H^\dagger$ almost surely. Therefore every convergent subsequence of $\hat{\theta}_H$ has the same limit $\theta_H^\dagger$, which proves

$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger.$$

This is exactly the conclusion of Newey and McFadden (1994, Theorem 2.1) specialized to the present criterion. □

Corollary 1 (Structural consistency under inception-only pricing)**.** *Suppose, in addition, that $H = 0$, that the pooled inception-only pricing restriction of Proposition 1 holds at the structural parameter θ_0, and that the inception-only population criterion $Q_0(\theta)$ is uniquely maximized at θ_0. Then*

$$\theta_0^\dagger = \theta_0 \qquad \textit{and hence} \qquad \hat{\theta}_0 \xrightarrow{p} \theta_0.$$

Corollary 1 isolates the special case in which the estimator converges to the structural SDF parameter. This separation is useful empirically because it prevents the reader from conflating consistency of the extremum estimator with structural validity of a horizon-averaged criterion.

C.4 Proof of asymptotic normality

Building on Proposition 2, the central limit theorem is centered at the horizon-specific target $\theta_H^\dagger$. The argument combines the extremum-estimator expansion of Newey and McFadden (1994, Theorem 3.1) with a central limit theorem for NED random fields under increasing-domain asymptotics; see Jenish and Prucha (2012, Theorem 4). The nonstandard feature is the vintage-year dependence of the score contributions.

Proof of Proposition 3. Define the sample score and sample Hessian by

$$S_{n,H}(\theta) := \nabla_\theta Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} s_i^{(H)}(\theta), \qquad \mathcal{H}_{n,H}(\theta) := \nabla_{\theta\theta} Q_{n,H}(\theta).$$

Because $\theta_H^\dagger$ lies in the interior of Θ and $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$ by Proposition 2, $\hat{\theta}_H$ is an interior maximizer with probability approaching one. Hence the first-order condition holds with probability approaching one:

$$S_{n,H}(\hat{\theta}_H) = 0.$$

Applying a mean-value expansion of $S_{n,H}(\cdot)$ around $\theta_H^\dagger$ yields

$$0 = S_{n,H}(\theta_H^\dagger) + \mathcal{H}_{n,H}(\tilde{\theta}_H)\big(\hat{\theta}_H - \theta_H^\dagger\big),$$

where $\tilde{\theta}_H$ lies on the line segment joining $\hat{\theta}_H$ and $\theta_H^\dagger$. Rearranging gives

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) = -\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1}\sqrt{n}\, S_{n,H}(\theta_H^\dagger). \tag{38}$$

We now analyze the two factors on the right-hand side.

Step 1: asymptotic distribution of the score. By definition,

$$\sqrt{n}\, S_{n,H}(\theta_H^\dagger) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n} s_i^{(H)}(\theta_H^\dagger).$$

Under the regularity conditions collected in this appendix, a central limit theorem holds for these score contributions. In the present application, the required weak dependence is generated by the vintage-year random field, and the maintained NED and mixing conditions are exactly those covered by Jenish and Prucha (2012, Theorem 4) under increasing-domain asymptotics. Therefore,

$$\sqrt{n}\, S_{n,H}(\theta_H^\dagger) \xrightarrow{d} \mathcal{N}(0, \Lambda_H),$$

where

$$\Lambda_H = \lim_{n\to\infty} n \cdot \mathrm{Var}\left[\nabla_\theta Q_{n,H}(\theta_H^\dagger)\right].$$

Step 2: convergence of the sample Hessian. Because $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$, the intermediate point $\tilde{\theta}_H$ also satisfies $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$. Hence, with probability approaching one, $\tilde{\theta}_H \in \mathcal{N}_H$. Assumption 8 then gives

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) - \nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} 0.$$

Since Q_H is twice continuously differentiable in a neighborhood of $\theta_H^\dagger$, continuity of its Hessian and $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$ imply

$$\nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} \nabla_{\theta\theta} Q_H(\theta_H^\dagger) = \mathcal{H}_H.$$

Therefore,

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) \xrightarrow{p} \mathcal{H}_H.$$

By assumption, $\mathcal{H}_H$ is nonsingular, so the continuous mapping theorem yields

$$\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1} \xrightarrow{p} \mathcal{H}_H^{-1}.$$

Step 3: conclusion. Substituting the previous two steps into Equation (38) and applying Slutsky's theorem gives

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}\left(0, \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}\right).$$

This is the extremum-estimator central limit theorem of Newey and McFadden (1994, Theorem 3.1), with the non-iid score CLT supplied by the increasing-domain NED random-field result of Jenish and Prucha (2012, Theorem 4). □

Corollary 2 (Structural asymptotic normality in the inception-only case)**.** *Under the additional conditions of Corollary 1,*

$$\sqrt{n}\left(\hat{\theta}_0 - \theta_0\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_0).$$

Remark 5 (Interpretation of the asymptotic center)**.** *For $H > 0$, Proposition 3 is centered at $\theta_H^\dagger$, not at θ_0. Therefore, large-sample inference in the horizon-averaged case pertains to the pseudo-true parameter defined by the criterion itself. Structural interpretation relative to θ_0 requires the target-shift analysis developed later in the paper.*

C.5 Inference details for score, Hessian, bandwidth, and SHAC implementation

This subsection records the implementation details omitted from the compressed main-text discussion in Section 2.3.

Score contributions. It is useful to define the score contribution of unit i explicitly as

$$s_i^{(H)}(\theta) := \nabla_\theta L\left(\bar{\epsilon}_i^{(H)}(\theta)\right) = L'\left(\bar{\epsilon}_i^{(H)}(\theta)\right) \cdot \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta). \tag{39}$$

For the quadratic loss $L(x) = x^2$, Equation (39) simplifies to

$$s_i^{(H)}(\theta) = 2\bar{\epsilon}_i^{(H)}(\theta)\nabla_\theta \bar{\epsilon}_i^{(H)}(\theta).$$

This explicit definition makes clear that the long-run covariance matrix Λ_H is the covariance of the criterion score, not the covariance of the pricing errors themselves.

Hessian estimator. A sample analogue of the Hessian is

$$\hat{\mathcal{H}}_H := \nabla_{\theta\theta} Q_{n,H}(\hat{\theta}_H) = -\frac{1}{n}\sum_{i=1}^{n} \nabla_{\theta\theta} L\left(\bar{\epsilon}_i^{(H)}(\hat{\theta}_H)\right). \tag{40}$$

When analytic derivatives are unavailable, the derivatives entering Equation (40) can be computed numerically. In practice, central finite differences are typically adequate, although automatic differentiation or quasi-Newton approximations may also be used. For a methodological paper, the exact numerical routine is secondary; what matters asymptotically is that $\hat{\mathcal{H}}_H$ consistently estimates $\mathcal{H}_H$. In our empirical applications, we use the central difference approximations of (Eu, 2017, Algorithm 2).

SHAC estimator of the long-run covariance matrix. Because the score contributions exhibit vintage-year dependence, a heteroskedasticity- and autocorrelation-consistent covariance estimator is required. We use the spatial HAC estimator of (Kim and Sun, 2011, Equation 2) adapted to the vintage-year metric in Equation (17), where $d_{i,j}$ is the vintage-year distance from Assumption 3 and $K(\cdot)$ is a symmetric kernel satisfying the standard SHAC regularity conditions.

A convenient choice is the Bartlett kernel

$$K_{BT}(x) = \max(0, 1 - |x|).$$

The associated bandwidth sequence b_V should satisfy

$$b_V \to \infty, \qquad \frac{b_V}{V} \to 0, \tag{41}$$

which is the standard condition ensuring consistent long-run covariance estimation under weak dependence. Equation (41) is a population-theoretic requirement. In empirical work, one may implement a finite truncation rule for b_V and assess robustness across reasonable bandwidth choices.

This distinction between theory and implementation is important. A fixed bandwidth may be useful in a finite sample and can be justified as an approximation when dependence becomes negligible beyond a small vintage-year lag. However, under generic weak dependence, consistency of SHAC is established under a diverging bandwidth sequence such as Equation (41).

Asymptotic covariance matrix, standard errors, and Wald tests. Building on Equation (18), the vector of asymptotic standard errors is

$$\mathrm{SE}(\hat{\theta}_H) = \sqrt{\mathrm{diag}\left[\widehat{\mathrm{Avar}}(\hat{\theta}_H)\right]}. \tag{42}$$

This presentation is algebraically equivalent to the usual sandwich formula, but it is more transparent because the $1/n$ scaling is stated directly at the covariance-matrix level.

For linear hypotheses of the form

$$\text{Hypothesis}_0 : \mathcal{R}\theta = r, \qquad \text{Hypothesis}_1 : \mathcal{R}\theta \neq r,$$

the Wald statistic is

$$W_H = (\mathcal{R}\hat{\theta}_H - r)^\top \left[\mathcal{R}\widehat{\mathrm{Avar}}(\hat{\theta}_H)\mathcal{R}^\top\right]^{-1} (\mathcal{R}\hat{\theta}_H - r), \tag{43}$$

which is asymptotically χ^2_q under Hypothesis_0, where q is the number of linear restrictions.